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Abstract

Urban green space is a recognised determinant of public health and well-being. Existing accessibility frameworks measure how much green space residents can reach, but typically treat all parks as interchangeable regardless of quality. This project proposes a *quality-weighted* extension to existing green accessibility indices by integrating two independent research outputs: the five-dimensional park well-being scores of Dietz et al. [2] and the GHS-grid spatial framework of ATGreen [1].

Four novel indices are defined and computed for **ten cities** spanning five continents — Amsterdam, Berlin, Madrid, Melbourne, Montreal, New York, Paris, Rome, Singapore and Washington DC — across a combined 3.30 million population grid cells at 90 m resolution: Quality-Weighted Exposure (QWE), Quality-Adjusted Minimum Distance (QAMD), Gravity-Model Accessibility (GMA), and Best Reachable Score (BEST). Park geometries and scores are extracted from local OpenStreetMap PBF files using `osmium`, mapped to the Global Human Settlement (GHS) population grid, and linked via vectorised haversine distance computation.

Quality-weighted coverage varies by an order of magnitude across the sample, from 99.6% of cells in Paris to 9.9% in Melbourne (QWE). The central finding of this multi-city extension is that *most of that variation is an artefact of administrative boundary definition rather than of green-space provision*: the OSM relation used to delimit each city ranges from a compact municipality (Paris, 49,447 cells) to an entire metropolitan region (Melbourne, 1,626,690 cells), and the share of a city’s grid occupied by scored parks alone explains the coverage ranking almost completely (Pearson $r = 0.93$). A boundary-independent measure — the share of *reachable* cells served within 15 minutes — compresses the range to 55.9–92.4% and is proposed as the appropriate basis for cross-city comparison. The environmental dimension is the weakest in five of the ten cities, more than any other dimension. All modelling assumptions — straight-line distances, uniform walking speed, a 3 km grid buffer, and within-city standardisation of the quality scores — are stated explicitly in Section 4.9. The pipeline is self-contained, runs without external APIs or databases, and processes a country-sized OSM extract on a machine with under 4 GB of RAM.

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I am grateful to my supervisor, Linus Dietz, for proposing the question that became this project and for steering it through the point at which the multi-city results stopped looking like a bug and started looking like the finding. His own work on health-promoting park scores, together with the ATGreen framework of Battiston and Schifanella, provided the two foundations this project builds on; both were released openly, and neither extension would have been possible otherwise.

I also thank the OpenStreetMap contributors whose volunteered mapping constitutes the entire empirical base of this report, and Geofabrik for maintaining and publishing the regional extracts used throughout.

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Nomenclature

All abbreviations and symbols used in this report are listed below in alphabetical order. Each term is also defined at the point of first use in the body of the report.

Abbreviations

API	Application Programming Interface
BCS	British Computer Society
BEST	Best Reachable Score (index defined in Equation (4.8))
CC BY	Creative Commons Attribution licence
CRS	Coordinate Reference System
CSV	Comma-Separated Values (file format)
DC	District of Columbia
EU	European Union
GB	Gigabyte
GDPR	General Data Protection Regulation
GHS	Global Human Settlement (layer)
GHSL	Global Human Settlement Layer
GHS-POP	Global Human Settlement population grid
GHS-UCDB	Global Human Settlement Urban Centre Database
GMA	Gravity-Model Accessibility (index defined in Equation (4.7))
IET	Institution of Engineering and Technology
JRC	Joint Research Centre (European Commission)
MB	Megabyte
ODbL	Open Database License
OSM	OpenStreetMap
OSRM	Open Source Routing Machine
PBF	Protocolbuffer Binary Format (OpenStreetMap file format)
QAMD	Quality-Adjusted Minimum Distance (index defined in Equation (4.6))
QWE	Quality-Weighted Exposure (index defined in Equation (4.5))
RAM	Random Access Memory
WHO	World Health Organization

Symbols

a_j	Park overlap area of destination cell j (hectares, ha)
A_i	Generic gravity-model accessibility of origin i (dimensionless)
d	Well-being dimension, $d \in \{\text{cultural, environmental, nature, physical, social}\}$
d_{ij}	Great-circle (haversine) distance from cell i to cell j (metres, m)
i	Origin grid cell

j	Destination (green) grid cell carrying a park overlap and quality score
O_j	Opportunity mass at destination j in the generic gravity model (hectares, ha)
q	Quality threshold applied in QAMD (z-score units; default $q = 0$)
r	Pearson product-moment correlation coefficient (dimensionless)
R	Mean radius of the Earth, 6 371 000 m
$s_d(j)$	Raw quality score of cell j in dimension d (z-score units)
$s_d^{(01)}(j)$	Min-max normalised quality score of cell j in dimension d (dimensionless, $\in [0, 1]$)
t_{ij}	Walking time from cell i to cell j (minutes, min)
T	Walking-time threshold defining reachability (minutes, min; default $T = 15$)
τ	Decay constant of the gravity model (minutes, min; default $\tau = 10$)

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Chapter 1

Introduction

Urban green spaces are widely recognised as critical infrastructure for public health. Access to parks is associated with reduced stress [16], increased physical activity [17], and improved mental well-being [15]. However, most spatial accessibility metrics treat all parks as equivalent: a barren field contributes as much as a richly-equipped park with sports facilities, cultural installations, and nature areas.

This project addresses that gap by combining two complementary research outputs into a single, self-contained pipeline:

- **Dietz et al. [2]** — A lexicon-based framework that scores parks across five health-promoting dimensions (cultural, environmental, nature, physical, social) using OpenStreetMap (OSM) element and space counts, normalised for park size via log-log regression residuals.
- **ATGreen [1]** — A multi-city green accessibility framework that uses the Global Human Settlement (GHS) population grid as its spatial base and computes three accessibility indices using street-network walking distances.

The core contribution is four *quality-weighted* accessibility indices that weight park contributions by their Dietz et al. quality scores. In outline, the method is as follows: each city’s boundary and its scored parks are read from a local OpenStreetMap extract; a ~ 90 m population-grid frame is laid over the city; each grid cell inherits an area-weighted quality score from the parks it overlaps; and walking-time distances between all cells then drive four indices that answer complementary questions — how much quality is reachable (QWE), how far the nearest above-average park is (QAMD), how access decays with distance (GMA), and what the best reachable park is (BEST). Chapters 3–4 develop each step. An earlier version of this work computed these indices for Amsterdam alone. This report extends the analysis to ten cities drawn from the 35-city joint study sample, which makes it possible to ask not only how accessible a single city is, but whether the indices behave consistently when the pipeline is applied across very different urban forms and, critically, across very different administrative boundary conventions.

Research questions:

1. What share of each city’s residential grid cells can reach a high-quality park within a 15-minute walk, broken down by well-being dimension?
2. How do quality-weighted indices differ from the baseline (unweighted) indices in terms of coverage and spatial distribution?
3. Which well-being dimensions show the lowest accessibility, and is the pattern consistent across cities?

4. Are raw coverage percentages comparable across cities, and if not, what measure is?

Chapter 2

Background and Related Work

2.1 ATGreen: Multi-Dimensional Green Accessibility

Green accessibility research sits at the intersection of two literatures. Public-health work establishes *why* access matters: proximity to green space is associated with reduced stress [16], higher physical activity [17] and better self-reported well-being [15], and the WHO recommends universal access within a short walk. Accessibility modelling establishes *how* to measure it, from Hansen’s original gravity formulation [6] through spatial-interaction models [7] to the positive/normative distinction drawn by Páez et al. [8], which this project follows in reporting descriptive indices rather than prescriptive targets. The gap between the two literatures is quality: health effects plausibly depend on what a park offers, yet accessibility measures have treated green space as homogeneous.

Battiston and Schifanella [1] argue that no single indicator is sufficient to characterise green accessibility in a city. Analysing over 1,000 cities worldwide, they demonstrate that rankings induced by minimum distance, exposure, and per-capita access are only weakly correlated, and that different segments of a city’s population may score high on one metric while scoring low on another. They propose a multi-dimensional framework with three baseline indices:

Minimum distance $\min_{j \in \text{green}} d_{ij}$ — the walking time from cell i to the nearest green cell.

Exposure $\sum_{j: d_{ij} \leq T} s_j$ — total green area reachable within time threshold T .

Per-person green area per capita within T , accounting for competing demand from co-located residents.

ATGreen uses the GHS-POP population raster [3] at 3 arc-second (≈ 90 m) resolution as its spatial unit, and computes walking distances via the Open Source Routing Machine (OSRM) [5].

2.2 Dietz et al.: Health-Promoting Park Scores

Dietz et al. [2] provide a replicable pipeline for scoring parks across five health-promoting activity dimensions. Starting from OSM PBF files, they:

1. Extract all OSM elements (nodes) and sub-areas (ways) within each park boundary.
2. Match tags to a hand-curated lexicon mapping key-value pairs to activity categories.
3. Compute raw counts of elements and space coverage per dimension.
4. Fit log-log regressions of activity against park area to obtain size-normalised residuals.

5. Standardise residuals to z-scores within each city.

6. Average the nodes z-score and spaces z-score to produce a `combined_score` per dimension.

The five dimensions and their interpretations are shown in Table 2.1.

Table 2.1: The five well-being dimensions of Dietz et al. [2].

Dimension	What it captures
Cultural	Cultural facilities (museums, theatres, monuments, art installations)
Environmental	Gardening and active conservation of nature (community gardens, habitat maintenance, conservation activity)
Nature	Biodiversity and nature (trees, gardens, natural areas, wildlife habitats)
Physical	Physical activity infrastructure (sports courts, playgrounds, running paths)
Social	Social interaction spaces (benches, picnic areas, cafes, community facilities)

2.3 Green Space Accessibility: Conceptual Foundations

2.3.1 Gravity Models

The gravity model of accessibility originates with Hansen [6], who proposed that the attraction of a destination decays with distance. The exponential decay form used here follows Wilson [7]:

$$A_i = \sum_j O_j \cdot f(d_{ij}), \quad f(d) = e^{-d/\tau}$$

where O_j is the opportunity mass at destination j and τ controls the rate of decay.

2.3.2 Walking Speed and Threshold Standards

A standard adult walking speed of 5 km/h (83.3 m/min) [12, 13] is used to convert distances in metres to minutes. The 15-minute threshold for green space access is formalised in the EU Biodiversity Strategy for 2030 [14], which targets all urban residents within 15 minutes' walk of a green space.

2.3.3 Areal Interpolation

Assigning park-level scores to grid cells when a cell overlaps multiple parks uses area-weighted averaging, a standard technique in spatial data conflation. The foundational reference is Goodchild and Lam [9], who formalise areal interpolation as a proportion-preserving mapping from source zones to target zones.

Chapter 3

Data

3.1 Study Area and Cities

The 35 cities shared by ATGreen and Dietz et al. span six continents and were originally selected from the Global Human Settlement Urban Centre Database (GHS-UCDB) [4], retaining those with at least 100,000 inhabitants and, per the upstream selection of Battiston and Schifanella [1], a usable OSM administrative boundary and green-space mapping adequate for index computation. No independent data-quality audit was performed for this project; the upstream selection is inherited as given.

Ten of these were processed for this report, chosen to span continents, urban forms and — as it turned out to matter a great deal — boundary conventions: Amsterdam, Berlin, Madrid, Melbourne, Montreal, New York, Paris, Rome, Singapore and Washington DC.

Each city is delimited by its OpenStreetMap administrative relation, and Table 3.1 lists the relation used together with the Geofabrik [19] extract it was read from. It is worth drawing attention at this early stage to the final column, because it conditions the interpretation of every result that follows: these relations are *not* equivalent units. Relation 71525 is the City of Paris within the Boulevard Périphérique; relation 4246124 is the whole of Greater Melbourne including its rural fringe. The former yields 49,447 grid cells, the latter 1,626,690 — a 33-fold difference that reflects administrative practice, not city size in any functional sense.

Table 3.1: OSM relations and source extracts for the ten cities processed.

City	Extract	Relation	Boundary covers
Paris	ile-de-france	71525	City of Paris (intra-muros)
Washington DC	district-of-columbia	162069	Federal district
Amsterdam	noord-holland	271110	Municipality
Montreal	quebec	1634158	City of Montreal (island)
Madrid	spain	5326784	Municipality
New York	new-york	175905	New York City (five boroughs)
Berlin	berlin	62422	City-state
Singapore	malaysia-singapore-brunei	536780	Whole country
Rome	centro	41485	Comune di Roma
Melbourne	victoria	4246124	Greater Melbourne metropolitan area

3.2 Park Quality Scores (Dietz et al.)

The pre-computed park quality scores used in this project are provided by the replication package of Dietz et al. [2] as two CSV files:

scores_cities.csv Combined quality scores (one row per park per city, five `combined_score_*` columns).

fingerprints_cities.csv Raw OSM element and space counts per park, including the `osm_id` field used to locate park geometries in the PBF file.

3.3 OpenStreetMap Data

3.3.1 OpenStreetMap as a Research Data Source

OpenStreetMap is the de facto standard source for park-level urban analysis at multi-city scale: it is the only geodata source with globally consistent tagging, park-level feature detail (benches, sports pitches, water features), and open licensing — the combination both ATGreen [1] and Dietz et al. [2] depend on, and no municipal open-data portal offers it across ten cities on five continents. Its limitation is equally well documented: coverage is produced by volunteers, and completeness varies systematically with place, being richer in affluent, well-populated areas [29]. Both properties matter here — the first makes the study possible, the second is carried through this report as a fairness caveat (Chapter 7).

3.3.2 Source Extracts

All geometry is read from local OpenStreetMap PBF extracts published by Geofabrik [19]; no Overpass or other API is queried at any point, so the pipeline is fully reproducible offline given the same files. Table 3.2 lists the ten extracts.

Table 3.2: OpenStreetMap PBF extracts used, with the OSM snapshot each represents.

City	Extract file	Size	Snapshot
Washington DC	district-of-columbia-260730	20 MB	2026-07-30
Berlin	berlin-260730	94 MB	2026-07-30
Amsterdam	noord-holland-260620	179 MB	2026-06-20
Melbourne	victoria-latest	228 MB	2026-07-27
Singapore	malaysia-singapore-brunei-260730	237 MB	2026-07-30
Paris	ile-de-france-260730	320 MB	2026-07-30
Rome	centro-260730	363 MB	2026-07-30
New York	new-york-260730	471 MB	2026-07-30
Montreal	quebec-latest	1,104 MB	2026-07-27
Madrid	spain-latest	1,397 MB	2026-07-27

The extracts differ by two orders of magnitude in size, and — importantly for the methodology — in the *kind* of region they cover. Berlin and Washington DC are published at city scale, so the extract is close to the study area. Others are only available at region scale (Île-de-France for Paris, Centro for Rome, Victoria for Melbourne, New York State for New York City) or at country scale (Spain for Madrid, the combined Malaysia–Singapore–Brunei file for Singapore, Quebec for Montreal). The city of interest is a small fraction of these larger files, which has direct consequences for extraction cost

and memory: resolving node coordinates naively across a 1.4 GB country extract exhausts memory on a modest machine, and Section 4.2 describes how this is avoided.

3.3.3 Temporal Consistency

The snapshots are not all from the same date. Eight fall within a five-day window (2026-07-27 to 2026-07-30), but the Amsterdam extract is from 2026-06-20 — carried over from the earlier single-city study so that its published results remain exactly reproducible. OSM changes continuously, so in principle a park mapped in July but not in June would be missing from Amsterdam alone.

In practice the effect is negligible at this scale: the quantity that matters is which of the target parks from the Dietz fingerprints are present, and a six-week gap is small relative to the multi-year gap between the Dietz data collection and any of these snapshots (Section 4.9). The temporal mismatch that does matter is the one between the score data and the geometry data, not the one between extracts.

3.3.4 City Boundaries

Each city boundary is reconstructed from its OSM administrative relation, listed in Table 3.1, by collecting the relation's member ways and assembling them into a polygon. The relations range from 47 member ways (Amsterdam) to 168 (Melbourne). As Section 5.3 shows, the fact that these relations follow no common convention — municipality, city-state, federal district, whole country, or entire metropolitan region — is the single most consequential property of the input data for interpreting the results.

3.3.5 Park Polygons

Parks are features tagged `leisure=park`, filtered to the OSM IDs listed for that city in the Dietz fingerprints file, and extracted as both simple ways and multipolygon relations. Features tagged `access=no` or `access=private` are excluded, as are `leisure=nature_reserve` and `landuse=recreation_ground`, which the Dietz scoring does not cover.

Across the ten cities, 7,459 of 7,910 target parks were located (94.3%), ranging from 1,787 parks in Melbourne down to 106 in Amsterdam. Recovery is discussed with the results in Table 5.1. Rather than leaving the shortfall to speculation, Amsterdam's 14 missing parks were inspected individually, and they are *not* negligible: they include Vondelpark (48.0 ha, the city's flagship park) and Lange Bretten (119.4 ha), together 24.4% of Amsterdam's scored park area. Tracing Vondelpark in the extract identifies the mechanism: its original way (OSM ID 44213014) still exists but now carries no tags, its attributes having migrated to a new multipolygon relation (17080648) that the fingerprints, keyed to the old ID, cannot match. The parks are present in OSM but unreachable through the ID-based join — identifier churn, not data loss. QWE and GMA values for Amsterdam are therefore understated, most severely near the missing large parks. For Washington DC the additional mechanism is different: a tightly clipped federal-district extract excludes parks lying just outside the boundary.

A small number of extracted parks have invalid geometry: four in total, one each in Singapore and New York and two in Paris, all arising from self-touching rings produced when multipolygon relations are assembled. These are repaired with a zero-width buffer before any spatial operation, the same correction the extraction code already applies to simple park ways.

3.4 Global Human Settlement Grid (GHS-POP)

The spatial base follows the GHS-POP grid *convention* [3]: 3 arc-second cells (≈ 90 m at the equator) aligned to the raster's coordinate origin. To be precise about what is consumed: *no GHSL raster file is read by the pipeline*. Population values are not used — a stated limitation — so only the grid geometry, cell size and origin, is reproduced, and cells are generated directly from those constants (Figure 4.1). The citation refers to the current R2023A release as the authoritative definition of that convention; the obsolescence notice attached to the withdrawn 2022 data package does not affect any input actually used. Alignment is preserved so GHS-POP population counts can later be joined per cell without resampling.

Chapter 4

Methodology

4.1 Pipeline Overview

The entire pipeline is implemented in `quality_accessibility_osmium.py`, a self-contained Python script with no external API calls or database requirements. It runs in six sequential steps followed by index computation:

1. Extract city administrative boundary from PBF
2. Load target park OSM IDs from Dietz fingerprints CSV
3. Extract park geometries from PBF
4. Generate GHS-aligned grid cells over the city
5. Compute park-cell spatial overlap fractions
6. Assign cell-level quality scores via area-weighted averaging
7. Compute four quality-weighted accessibility indices

4.2 Boundary and Park Extraction

City boundaries and park polygons are both reconstructed from the OSM extract: the boundary from the city’s administrative relation, and parks from the `leisure=park` features whose OSM IDs appear in the Dietz fingerprints. Multipolygon parks (those with holes or multiple parts) are assembled following the standard OSM multipolygon convention [20]. Features tagged `access=no` or `access=private` are excluded. Implementation detail — the two-pass streaming pattern over the PBF, and a targeted variant that resolves coordinates only for the known park IDs so that country-sized extracts (Spain, 1.4GB) fit in a few GB of memory — is documented with the code rather than here; both variants were verified to produce identical geometry.

One aspect does warrant reporting because it affects results: four parks across Singapore, Paris and New York carried invalid self-touching geometry after assembly and are repaired with a zero-width buffer before use, the same correction applied to simple park ways. Without it, the overlap computation fails for those three cities entirely.

Pipeline stages and the artefact each produces

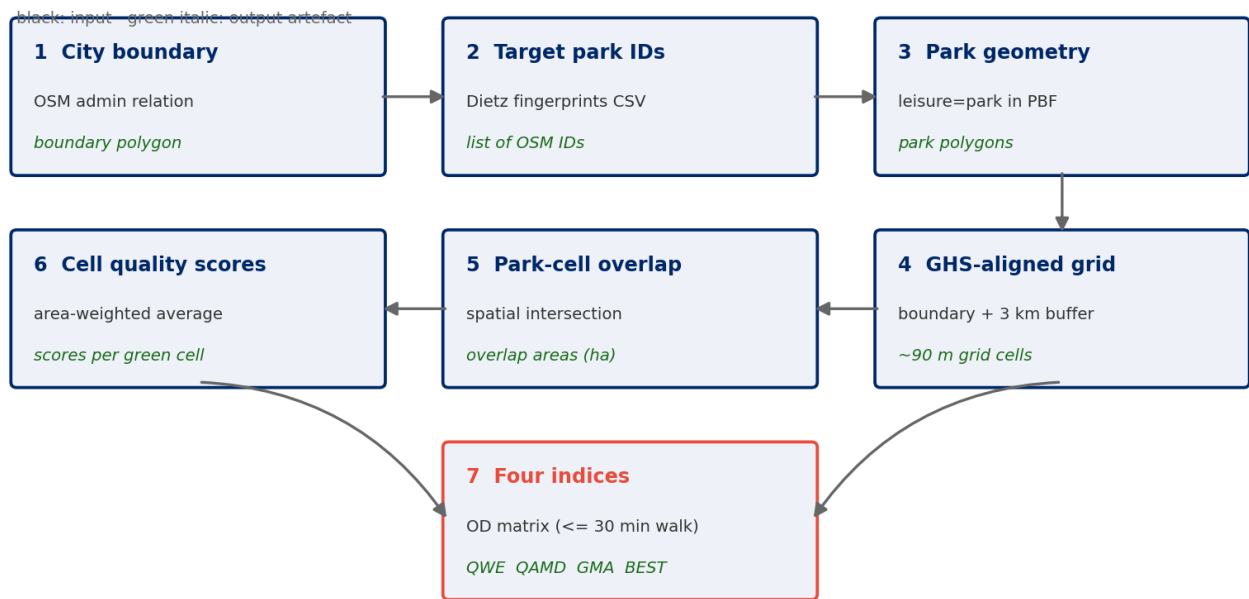


Figure 4.1: The seven pipeline stages, with the input each consumes and the artefact each produces. Stages 1–3 read OpenStreetMap; stages 4–6 build the spatial frame and attach quality; stage 7 computes the indices.

4.3 GHS Grid Generation

Grid cells are generated to match the GHS-POP raster convention rather than read from a raster file: no population values are used in this study, so the only thing required from GHS-POP is its geometry. Cells of 3 arc-seconds are laid out from the raster’s coordinate origin across the city boundary plus a 3 km buffer, and cells not intersecting the buffered boundary are discarded. Because the cells coincide exactly with GHS-POP pixels, a future population-weighted analysis can join population counts per cell without resampling.

4.4 Park-Cell Spatial Overlap

The spatial join between parks and cells is performed in the local UTM CRS (metric coordinates) using `geopandas.sjoin`. For each (cell, park) pair that intersects, the actual intersection area in hectares is computed:

$$a_{ij} = \text{Area}(\text{cell}_i \cap \text{park}_j) \quad [\text{ha}]$$

This extends ATGreen’s `osm2grid` remapping step by preserving both *which* park overlaps each cell and *how much* — information required for quality-weighting. The result is a table of 2,492 (cell, park) intersection pairs.

4.5 Cell-Level Quality Score Assignment

Park scores from Dietz et al. are assigned to grid cells via area-weighted averaging [9]. For each green cell i and quality dimension d :

$$\hat{s}_{i,d} = \frac{\sum_j a_{ij} s_{j,d}}{\sum_j a_{ij}} \quad (4.1)$$

where j iterates over all parks overlapping cell i , a_{ij} is the overlap area in hectares, and $s_{j,d}$ is the Dietz combined score for park j in dimension d .

To ensure non-negative contributions in QWE and GMA (where negative scores would incorrectly penalise accessibility), scores are also provided in a min-max normalised form:

$$\hat{s}_{i,d}^{(01)} = \frac{\hat{s}_{i,d} - \min_i \hat{s}_{i,d}}{\max_i \hat{s}_{i,d} - \min_i \hat{s}_{i,d}} \quad (4.2)$$

Of the 2,492 overlap pairs, 2,424 green cells received quality scores after joining with the Dietz data.

4.6 Distance Computation

4.6.1 Haversine Geodesic Fallback

In the absence of a local OSRM server, cell-to-cell distances are computed using the haversine great-circle formula [10]:

$$d = 2R \arcsin\left(\sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_1 \cos \phi_2 \sin^2 \frac{\Delta\lambda}{2}}\right) \quad (4.3)$$

where ϕ denotes latitude, λ longitude, and $R = 6,371,000$ m is Earth's mean radius. Walking minutes are computed at 5 km/h:

$$t_{ij} = \frac{d_{ij}}{5000/60} \quad [\text{min}] \quad (4.4)$$

The computation is vectorised with a spatial pre-filter, and only origin–destination pairs within 30 minutes' walk are retained; Amsterdam yields 9,072,560 such pairs in under two minutes on a standard laptop. Implementation specifics are left to the code.

4.6.2 Consequence of Geodesic Rather than Street-Network Distances

Straight-line distances understate real walking routes by typically 20–30%, so every coverage figure in this report is an *optimistic bound*: a 15-minute geodesic threshold corresponds roughly to an 11–12-minute threshold on the street network. The bias is also uneven between cities — largest where the pedestrian network is least permeable (large-block grids, waterways severing routes) and smallest in dense European cores — so cross-city gaps are slightly compressed or exaggerated depending on network form. Because the index functions are column-agnostic, replacing geodesic minutes with OSRM [5] walking minutes requires no change to any index computation, and this is the most direct upgrade path.

4.7 Quality-Weighted Accessibility Indices

This section defines the four measures that constitute the contribution of the project. Each answers a different question a resident or planner might ask — how much quality is nearby, how far the nearest good park is, how access decays with distance, and what the best reachable option is — and each is computed separately for the five well-being dimensions, so no information is lost to averaging. Four indices are computed. All are expressed as functions of:

- i — origin grid cell
- j — green destination cell (has a park overlap and a quality score)
- d — quality dimension $\in \{\text{cultural, environmental, nature, physical, social}\}$
- a_j — park overlap area of cell j in hectares
- $s_d(j)$ — quality score of cell j in dimension d
- t_{ij} — walking time in minutes from i to j

4.7.1 Quality-Weighted Exposure (QWE)

Extends: ATGreen `exposure_index()` [1].

ATGreen baseline:

$$\text{Exposure}_i = \sum_{j: t_{ij} \leq T} a_j$$

QWE extension:

$$\text{QWE}_i^d = \sum_{j: t_{ij} \leq T} a_j \cdot s_d^{(01)}(j) \quad (4.5)$$

Normalised scores $s_d^{(01)}$ (Equation (4.2)) are used so that all parks contribute positively. The threshold $T = 15$ minutes operationalises the EU Biodiversity Strategy target [14]. A larger QWE indicates that more high-quality green space is reachable — a low-quality park covering the same area contributes proportionally less.

4.7.2 Quality-Adjusted Minimum Distance (QAMD)

Extends: ATGreen `minimum_distance_index()` [1].

ATGreen baseline:

$$\text{MinDist}_i = \min_{j \in \text{green}} t_{ij}$$

QAMD extension:

$$\text{QAMD}_i^d(q) = \min_{j: s_d(j) \geq q} t_{ij} \quad (4.6)$$

Only parks whose quality score meets threshold q are considered. With raw z-scores and $q = 0$ (the default), only above-average parks qualify. A result of NaN indicates no qualifying park is reachable within the 30-minute search radius.

4.7.3 Gravity-Model Accessibility (GMA)

Inspired by: Hansen [6], Wilson [7].

$$\text{GMA}_i^d = \sum_j a_j \cdot s_d^{(01)}(j) \cdot e^{-t_{ij}/\tau} \quad (4.7)$$

Unlike QWE, which applies a binary distance cutoff, GMA applies a continuous exponential decay with half-life $\tau = 10$ minutes. Example decay values: a park at 5 min contributes $e^{-0.5} \approx 0.61$ of its face value; at 15 min, $e^{-1.5} \approx 0.22$; at 30 min, $e^{-3} \approx 0.05$.

4.7.4 Best Reachable Score (BEST)

Novel index (no prior equivalent in ATGreen or Dietz).

$$\text{BEST}_i^d(T) = \max_{j: t_{ij} \leq T} s_d(j) \quad (4.8)$$

This is computed for $T \in \{5, 15, 30\}$ minutes, yielding 15 output columns (5 dimensions $\times$ 3 thresholds). The rationale is that satisfying a need for sport or cultural activity typically requires only one qualifying park. BEST answers: “*does at least one great park exist within a short walk?*” rather than “*how much total quality green space is reachable?*”

Table 4.1 summarises the four indices.

Table 4.1: Comparison of the four quality-weighted accessibility indices.

Index	Formula	Aggregation	What it answers
QWE	$\sum_{j: t_{ij} \leq T} a_j s_d(j)$	Sum within threshold	How much quality green exposure is reachable in total?
QAMD	$\min_{j: s_d(j) \geq q} t_{ij}$	Min over qualifying	How far is the nearest above-average park?
GMA	$\sum_j a_j s_d(j) e^{-t_{ij}/\tau}$	Gravity-weighted sum	Cumulative access weighted by distance decay
BEST	$\max_{j: t_{ij} \leq T} s_d(j)$	Max within threshold	Does at least one great park exist nearby?

4.8 Provenance of the Park Quality Scores

The five-dimensional park scores are used exactly as published in the Dietz et al. replication package [2]; they are an input to this project, not a contribution of it. Their construction — lexicon-based fingerprinting, log-log size normalisation, within-city z-standardisation — is summarised in Chapter 2 and described fully in the source publication. The one property essential to interpreting the results here is the within-city standardisation: “above average” always means above that city’s own mean, a point developed in Section 4.9.

4.9 Assumptions and Scope

The indices defined above rest on a number of modelling choices. They are collected here so that the results in the following chapter can be read against them.

4.9.1 Spatial Assumptions

Cell as point

Each 3 arc-second cell (≈ 90 m) is represented by its centroid for all distance computation.

Within-cell variation in where a resident actually lives is ignored, giving a positional error of up to roughly 60 m, or about 45 seconds of walking.

Grid extent

The grid covers the city’s administrative boundary *plus a 3 km buffer* (`buffer_m = 3000`), a value inherited from ATGreen, whose data-preparation notebooks set `BUFFER_SIZE=3000` for the same purpose: capturing parks just outside the boundary that are reachable from cells inside it. Every coverage percentage in this report therefore has a denominator larger than the city itself. Table 4.2 quantifies this: between 19% and 62% of the denominator lies outside the administrative boundary, and the effect is strongly scale-dependent — it inflates the denominator of a small, compact city such as Paris far more than that of a sprawling one such as Melbourne.

Equal cell weighting

Every cell counts equally regardless of how many people live in it. An uninhabited industrial or agricultural cell contributes to the denominator exactly as much as a dense residential block. The grid is GHS-aligned, so population weighting is available in principle but is not applied here.

City-local parks only

Only parks listed for that city in the Dietz fingerprints are extracted. A park immediately across a municipal border contributes nothing, so cells near the boundary have their accessibility systematically understated. This edge effect is largest for the smallest boundaries.

Table 4.2: Effect of the 3 km grid buffer on the denominator of every coverage figure.

City	Admin area (km ²)	+3 km buffer (km ²)	% of grid outside city
Paris	105.3	276.8	62
Washington DC	177.2	390.8	55
Amsterdam	219.4	483.2	55
Montreal	440.8	905.3	51
Madrid	603.9	1,080.4	44
Berlin	890.7	1,446.2	38
Rome	1,285.9	2,027.1	37
Singapore	1,638.4	2,425.1	32
New York	1,218.7	1,763.4	31
Melbourne	8,889.6	10,996.7	19

4.9.2 Travel Assumptions

Straight-line distance

Walking time is derived from geodesic (haversine) distance, not street-network routing. Real routes are typically 20–30% longer, so all coverage figures are optimistic bounds, and the bias is uneven — larger where the pedestrian network is less permeable.

Uniform walking speed

A constant 5 km/h is assumed for everyone. Gradient, crossings, weather, age and mobility impairment are not modelled. For a framework motivated by public health this is a strong simplification, since the populations most dependent on nearby green space are often the slowest walkers.

30-minute search radius

Origin–destination pairs beyond 30 minutes are discarded (`max_minutes = 30`). A cell whose nearest qualifying park is 35 minutes away is recorded as unreachable rather than distant, so QAMD is right-censored at 30 minutes.

15-minute threshold

The headline accessibility standard, used for QWE and for the QAMD percentages, is 15 minutes — the widely used “15-minute city” convention rather than a value derived from these data.

4.9.3 Index Parameters

All four indices depend on parameters fixed at their defaults rather than calibrated: the GMA distance-decay constant $\tau = 10$ minutes, the QAMD quality threshold $q = 0$ (a park qualifies if its z-score is at or above the city mean), and the BEST thresholds of 5, 15 and 30 minutes. QWE weights each park’s contribution by its overlap area in hectares multiplied by its quality score, so a large mediocre park can outweigh a small excellent one. No sensitivity analysis over these values was performed; the reported coverage figures would move if they were varied.

4.9.4 Quality Score Assumptions**Within-city standardisation**

This is the most consequential assumption for a multi-city study. The Dietz scores are z-standardised *within each city*, so “above average” in QAMD means above average *for that city*. A city whose parks are uniformly poor by international standards still has roughly half of them scoring above zero. QAMD therefore measures access to a city’s better parks, not access to parks meeting any absolute standard, and cross-city QAMD comparisons compare each city against its own baseline. Absolute cross-city quality claims cannot be made from these indices.

Tag-derived quality

Park quality is inferred entirely from OSM tag counts via the Dietz lexicon. Amenities that are unmapped, or mapped inconsistently between countries, are invisible. Mapping density varies systematically across the ten cities.

Temporal mismatch

The quality scores come from the Dietz collection, while the geometries come from 2026 OSM extracts. Parks created, deleted or re-tagged in the interval appear as extraction failures; 5.7% of target parks were not found overall, and 28.3% in Washington DC.

Park definition

Only features tagged `leisure=park` are considered, excluding `leisure=nature_reserve`, `landuse=recreation_ground` and informal green space. Features tagged `access=no` or `access=private` are excluded.

Area-weighted assignment

Where several parks overlap one cell, the cell’s score is the area-weighted mean of theirs, which smooths over the case of a cell adjacent to one excellent and one poor park.

Chapter 5

Results

5.1 Extraction Summary

Table 5.1 summarises the pipeline extraction for all ten cities. Across the sample, 7,459 of 7,910 target parks were located in the OSM extracts (94.3%), and 3,299,545 grid cells were generated in total.

Table 5.1: Pipeline extraction summary for the ten cities. “Target” is the number of parks listed for that city in the Dietz fingerprints; “Found” is the number located in the PBF extract.

City	Target	Found	%	Grid cells	Overlap pairs	Green cells
Paris	1,553	1,501	96.7	49,447	8,861	8,351
Washington DC	339	243	71.7	59,107	3,215	2,990
Amsterdam	120	106	88.3	92,655	2,492	2,424
Montreal	1,076	1,009	93.8	151,468	9,938	9,585
Madrid	457	428	93.7	166,559	7,527	7,169
New York	1,162	1,089	93.7	272,050	21,302	20,742
Berlin	571	545	95.4	277,295	9,361	9,116
Singapore	441	427	96.8	285,873	5,593	5,358
Rome	364	324	89.0	318,401	8,717	8,450
Melbourne	1,827	1,787	97.8	1,626,690	22,853	21,691
Total	7,910	7,459	94.3	3,299,545	99,859	95,876

Park recovery is high and consistent (89–98%) with one clear exception: Washington DC, where only 71.7% of target parks were located. The District of Columbia extract is the smallest in the sample (20 MB) and is tightly clipped to the federal district, so parks recorded in the Dietz fingerprints that sit just outside the district boundary — or that have been re-tagged or deleted from OSM since the Dietz data collection — are absent. The same mechanism accounts for the 11.7% shortfall in Amsterdam and 11.0% in Rome.

5.2 Index Coverage Across Cities

Table 5.2 reports, for each city, the share of its *own* grid cells reached by each index. Figure 5.1 presents the same information graphically.

The spread is extreme: QWE coverage ranges from 99.6% in Paris to 9.9% in Melbourne, a factor of ten. The ordering of the three indices is nevertheless identical in every city — GMA > QWE > QAMD₁₅ —

Table 5.2: Accessibility coverage as a percentage of each city’s own grid. QWE and GMA count cells with a non-zero value in any dimension; QAMD 15 min is the mean across the five dimensions of cells reaching an above-average park within 15 minutes.

City	Grid cells	QWE %	GMA %	QAMD 15 min %
Paris	49,447	99.6	100.0	91.7
Montreal	151,468	61.8	80.3	49.3
New York	272,050	57.0	72.4	37.9
Washington DC	59,107	52.0	73.8	34.5
Berlin	277,295	47.8	66.1	33.1
Amsterdam	92,655	32.4	50.6	23.8
Madrid	166,559	27.5	37.8	20.6
Rome	318,401	25.5	38.8	16.1
Singapore	285,873	20.6	28.0	15.5
Melbourne	1,626,690	9.9	12.1	6.7

which is the expected behaviour and a useful internal consistency check. GMA is the most permissive because its exponential decay imposes no hard cutoff; QAMD at 15 minutes is the most demanding because it requires an *above-average* park within a fixed walking time.

Amsterdam, the original case study, sits sixth of ten on every index. Its figures reproduce the earlier single-city report exactly (QWE 32.4%, GMA 50.6%), confirming that the multi-city pipeline did not perturb the original result.

Figure 5.2 places the two extremes side by side.

5.3 Coverage is Driven by Boundary Definition

The headline ranking in Table 5.2 is not a ranking of green-space provision. It is very close to a ranking of how tightly each city’s OSM administrative relation is drawn around its dense core.

Table 5.3 makes the mechanism explicit. Paris contains 1,501 scored parks within 49,447 cells — 30.4 parks per thousand cells, and 16.9% of the grid is occupied by a scored park. Melbourne contains more parks in absolute terms (1,787) but spreads them across 1,626,690 cells, giving 1.1 parks per thousand cells and a green-cell share of 1.3%. The Melbourne relation includes a large rural periphery in which almost no cell is within walking distance of anything.

Across the ten cities, the green-cell share and QWE coverage correlate at Pearson $r = 0.93$. In other words, roughly 86% of the variance in the headline coverage figure is explained by a single quantity — how much of the delimited area happens to be park — which is itself a direct consequence of where the administrative boundary was drawn. Any cross-city claim resting on Table 5.2 is therefore largely a claim about administrative geography.

5.4 A Boundary-Independent Comparison

A measure that conditions on the cells capable of reaching a park at all removes most of this distortion. Table 5.4 reports, for cells that reach an above-average park within the 30-minute search radius, the percentage that do so within 15 minutes. Cells in the rural fringe, which reach nothing, no longer dilute the denominator.

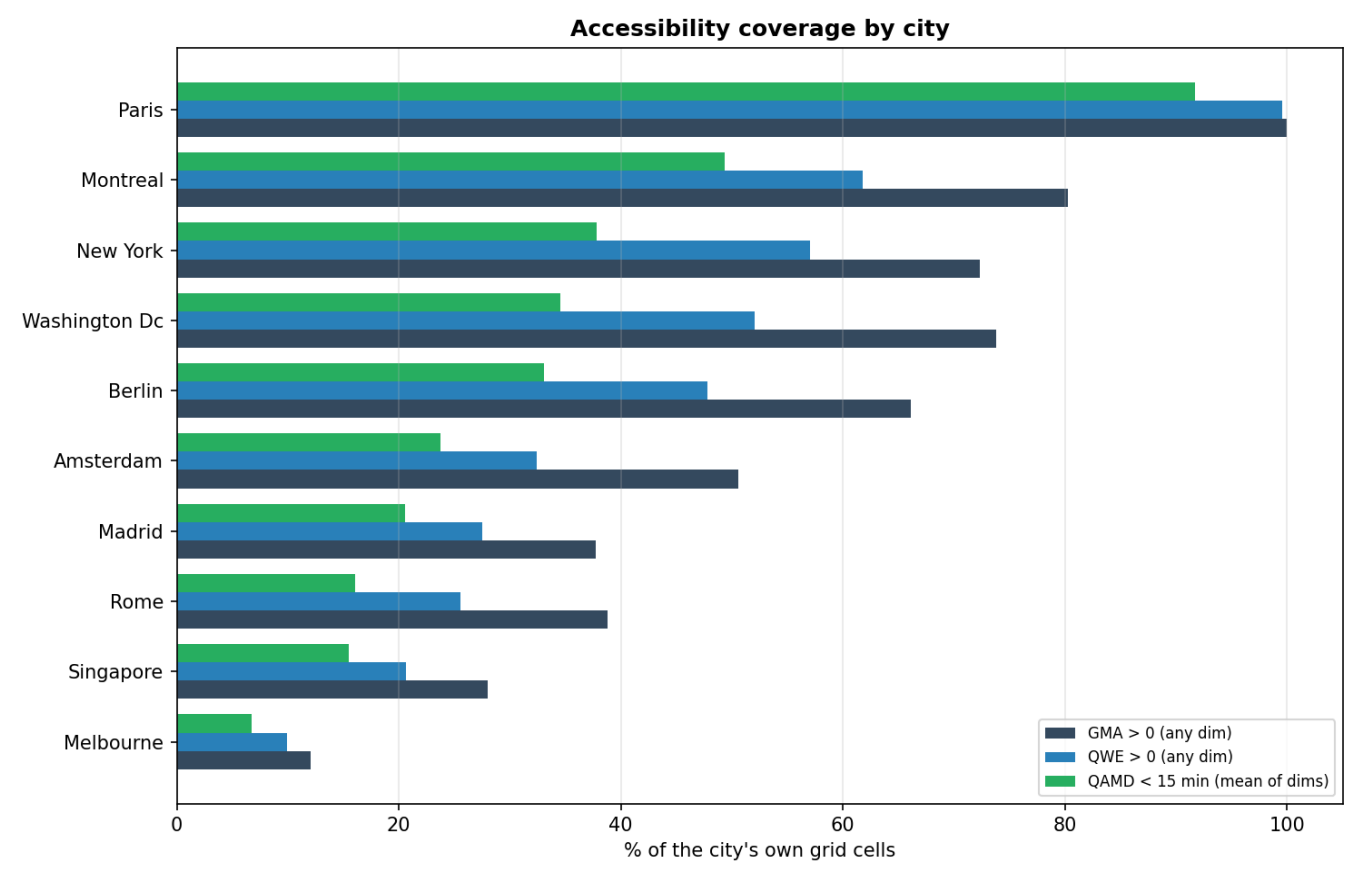


Figure 5.1: Accessibility coverage by city and index, as a percentage of each city’s own grid cells.

The effect is substantial. The range narrows from 9.9–99.6% to 55.9–92.4%, and the ordering changes materially: Melbourne rises from last place to third, while Amsterdam falls from sixth to ninth. Melbourne’s parks, once reached, are well distributed — it is the extent of unserved periphery inside its boundary, not the quality of its park network, that produced the 9.9% figure. Paris remains first on either measure, which is the one robust ranking claim the data supports.

Figure 5.3 shows the same quantity as a heatmap over all city–dimension pairs, expressed here as a share of all cells.

5.5 Dimensional Patterns

Table 5.5 records, for each city, the best- and worst-served dimension and the gap between them.

Two patterns hold across the sample. First, **the environmental dimension is the worst-served in five of the ten cities** (Amsterdam, Madrid, Melbourne, New York, Paris) — more often than any other dimension, and never the best in any city. The finding reported for Amsterdam in the single-city study therefore generalises: parks scoring highly for gardening and active nature conservation are scarce almost everywhere in this sample. Second, **the physical dimension is the best-served in five cities** and is never the worst, consistent with sports pitches and playgrounds being the most routinely provided park amenity.

The gap itself, plotted for all ten cities in Figure 5.4, varies widely, from 6.1 percentage points in Berlin to 38.0 in New York. Berlin, Singapore and Melbourne have relatively even dimensional profiles; New York and Montreal are markedly uneven. A city can therefore be well served on average while leaving one dimension badly under-provided — which is precisely the argument for a multi-dimensional

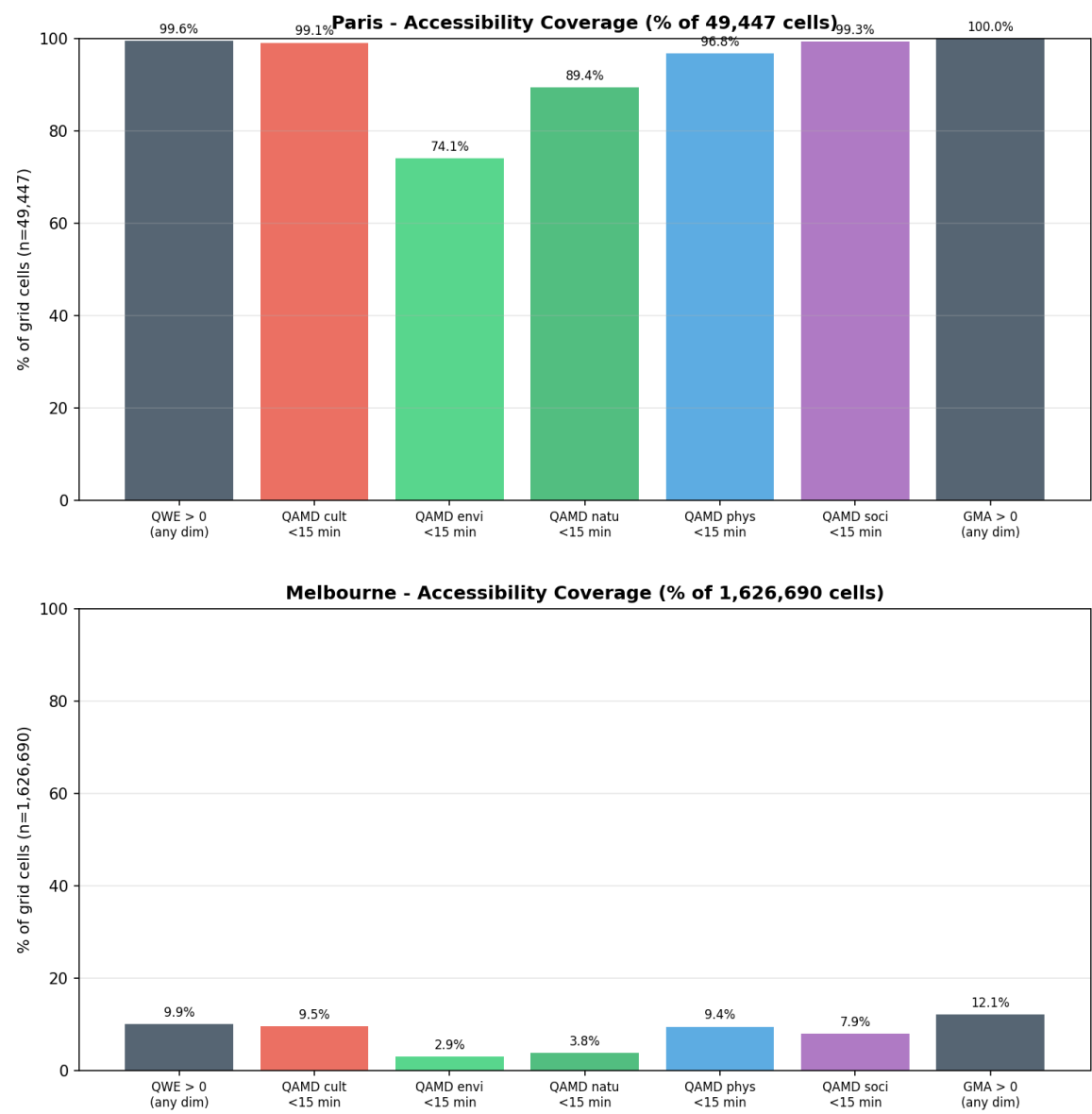


Figure 5.2: The two extremes of the sample. Paris (top): 49,447 cells, near-total coverage. Melbourne (bottom): 1,626,690 cells, coverage below 13% on every index. Section 5.3 argues this contrast is largely an artefact of boundary definition.

framework rather than a single green-accessibility indicator.

Figure 5.5 shows mean QWE intensity, which measures not whether a park is reachable but how much quality-weighted exposure a served cell receives.

Paris leads on intensity as well as coverage (means of 22.7–29.5 across dimensions). New York is a more interesting case: its nature and physical means (22.9 and 24.4) approach Paris’s while its cultural and environmental means (12.0 and 9.4) are roughly half, reflecting a park system with a small number of very large, nature-rich parks alongside many minimally-equipped ones.

5.6 Independence of the Five Dimensions

A premise of the multi-dimensional design is that the five dimensions measure genuinely different things. Figure 5.6 verifies this on the full 35-city Dietz score set: off-diagonal Pearson correlations

Table 5.3: Park density against coverage. “Green-cell share” is the percentage of the city’s grid cells that intersect a scored park.

City	Parks	Grid cells	Parks/1k cells	Green-cell %	QWE %
Paris	1,501	49,447	30.36	16.9	99.6
Montreal	1,009	151,468	6.66	6.3	61.8
Washington DC	243	59,107	4.11	5.1	52.0
New York	1,089	272,050	4.00	7.6	57.0
Madrid	428	166,559	2.57	4.3	27.5
Berlin	545	277,295	1.97	3.3	47.8
Singapore	427	285,873	1.49	1.9	20.6
Amsterdam	106	92,655	1.14	2.6	32.4
Melbourne	1,787	1,626,690	1.10	1.3	9.9
Rome	324	318,401	1.02	2.7	25.5

Table 5.4: Percentage of *reached* cells served within 15 minutes, by dimension. This conditions on cells that reach an above-average park at all, and is therefore far less sensitive to how the city boundary was drawn.

City	Cult.	Env.	Nature	Phys.	Social	Mean
Paris	99.1	76.0	90.7	96.8	99.3	92.4
Montreal	74.0	70.6	52.8	74.1	71.0	68.5
Melbourne	81.3	45.6	53.0	80.9	74.9	67.1
New York	70.8	50.0	68.6	74.3	63.7	65.5
Madrid	69.3	53.9	65.0	71.9	65.0	65.0
Singapore	66.2	69.6	66.7	65.4	56.1	64.8
Washington DC	63.1	56.8	60.1	68.1	60.8	61.8
Berlin	61.6	61.8	58.9	60.4	65.6	61.7
Amsterdam	56.5	48.8	60.0	62.2	60.6	57.6
Rome	56.8	57.2	55.4	59.6	50.6	55.9

between dimension scores range from $r = 0.09$ (environmental–physical) to $r = 0.31$ (nature–social). Scores this weakly correlated cannot be summarised by a composite without discarding most of their information, which is the empirical justification for reporting five dimensions separately throughout this report.

5.7 Spatial Pattern Within the City

Figure 5.7 maps total QWE per grid cell for Amsterdam, with the administrative boundary overlaid. Quality-weighted access is concentrated in the urban core and decays sharply outward: every cell within 4km of the centre has non-zero QWE, falling to 73% at 4–6km, 37% at 6–8km and 15% beyond 8km. Mean QWE peaks not in the historic centre but in the 2–4km ring (70.2 against 31.9 within 2km), reflecting the large high-scoring parks that ring rather than occupy the core.

5.8 Key Findings

- **Raw coverage is not comparable across cities.** QWE coverage spans 9.9–99.6%, but correlates at $r = 0.93$ with the share of the grid occupied by parks, which is a function of where

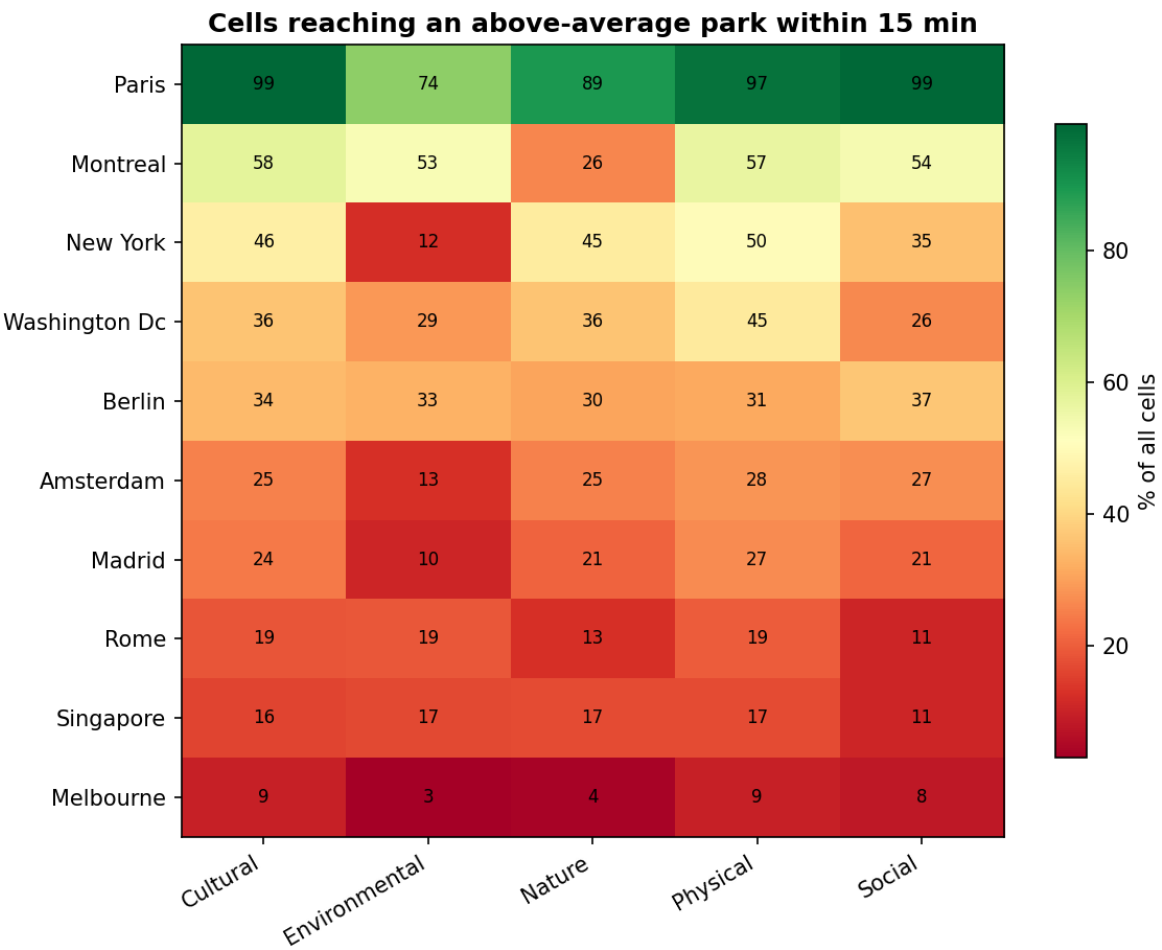


Figure 5.3: Cells reaching an above-average park within 15 minutes, by city and dimension, as a percentage of all cells.

the administrative boundary was drawn. Table 5.2 should not be read as a league table.

- **Conditioning on reachable cells largely fixes this.** The range narrows to 55.9–92.4% and the ranking changes substantially, with Melbourne rising from tenth to third.
- **Index ordering is stable.** $GMA > QWE > QAMD_{15}$ holds in all ten cities, as the index definitions predict.
- **Environmental accessibility is the weakest dimension overall**, worst-served in five of ten cities and best-served in none. The Amsterdam finding generalises.
- **Physical accessibility is the strongest**, best-served in five cities and worst in none — though Section 5.9 shows this is threshold-dependent, unlike the environmental finding.
- **Dimensional inequality varies far more than average provision.** The best–worst gap ranges from 6.1 points (Berlin) to 38.0 (New York), supporting the case for multi-dimensional measurement.
- **Amsterdam’s original results are reproduced exactly** (QWE 30,004 cells / 32.4%; GMA 46,860 / 50.6%; all five QAMD counts and 15-minute percentages), confirming the multi-city extension did not alter the single-city baseline.

Table 5.5: Best- and worst-served dimension per city, measured as the percentage of all cells reaching an above-average park within 15 minutes.

City	Best dimension	%	Worst dimension	%	Gap
New York	physical	50.2	environmental	12.2	38.0
Montreal	cultural	57.6	nature	25.9	31.7
Paris	social	99.3	environmental	74.1	25.3
Washington DC	physical	45.0	social	26.5	18.5
Madrid	physical	26.6	environmental	10.4	16.2
Amsterdam	physical	28.5	environmental	12.5	15.9
Rome	physical	19.5	social	10.7	8.8
Melbourne	cultural	9.5	environmental	2.9	6.6
Singapore	nature	17.0	social	10.7	6.3
Berlin	social	36.6	nature	30.5	6.1

5.9 Sensitivity Analysis

Section 4.9 notes that four parameters are fixed by convention rather than calibrated. This section varies each and reports how far the conclusions move. At their default values every sweep reproduces the published figures exactly, which also serves as an independent check on the results chapter.

5.9.1 Walking-Time Threshold

Table 5.6 varies the threshold T from 5 to 30 minutes. Absolute coverage naturally rises with T , but the ordering of the cities is almost perfectly preserved: the Spearman rank correlation against the $T = 15$ ordering is 0.976 at $T = 5$, 0.988 at $T = 10$ and exactly 1.000 at both $T = 20$ and $T = 30$.

Table 5.6: Mean QAMD coverage (% of all cells, averaged over the five dimensions) as the walking-time threshold varies. Cities are ordered by the published $T = 15$ column.

City	5 min	10 min	15 min	20 min	30 min
Paris	61.8	82.8	91.7	95.9	99.2
Montreal	22.2	38.1	49.3	57.5	70.7
New York	16.7	29.1	37.9	44.5	55.7
Washington DC	14.2	25.4	34.5	42.0	55.6
Berlin	12.1	23.4	33.1	41.1	53.6
Amsterdam	8.6	16.8	23.8	29.9	40.7
Madrid	10.1	16.1	20.6	24.3	31.1
Rome	5.8	11.2	16.1	20.4	28.5
Singapore	6.0	11.3	15.5	18.6	23.7
Melbourne	3.2	5.4	6.7	7.7	9.5

The dimensional conclusions are also stable. Environmental is the worst-served dimension in five of the ten cities at every threshold from 10 to 30 minutes, and in six at $T = 5$; only Berlin changes at all, switching from environmental to nature above 5 minutes. Every other city keeps the same worst-served dimension throughout. The claim that environmental accessibility is the weakest dimension is therefore not an artefact of the 15-minute convention.

The best-served dimension is less robust. Physical leads in seven cities at $T = 5$ but only five at

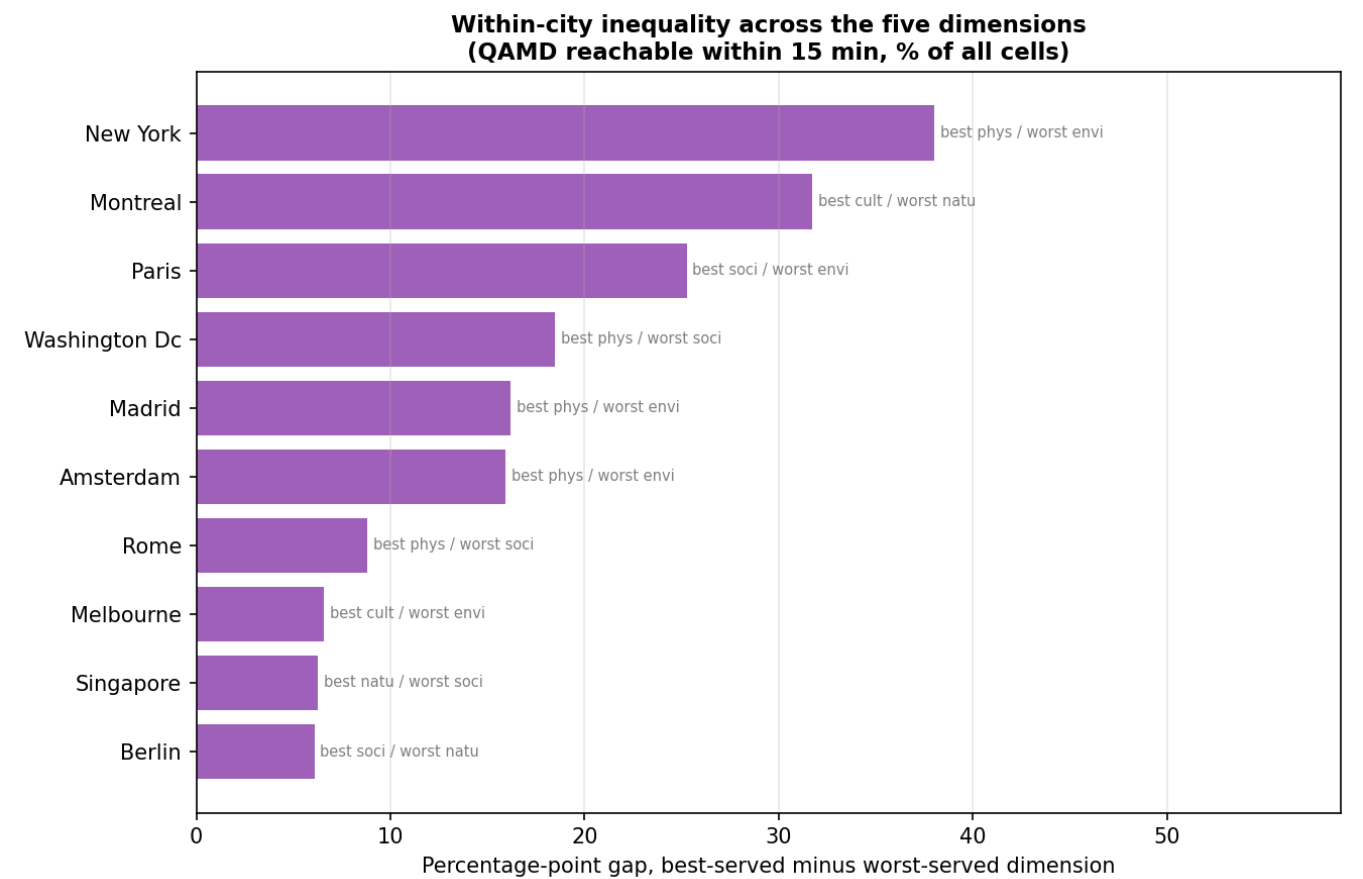


Figure 5.4: Within-city inequality across the five dimensions: the percentage-point gap between the best- and worst-served dimension.

$T \geq 15$, with Melbourne, Montreal, Paris, Rome and Singapore each changing as the threshold widens. The “physical is best-served” claim should accordingly be stated as threshold-dependent, unlike its environmental counterpart.

5.9.2 Grid Buffer

The 3km buffer is the parameter with by far the largest influence on the headline numbers. Table 5.7 recomputes every coverage figure using only cells whose centroid lies inside the administrative boundary.

The levels move enormously. Amsterdam’s headline QWE coverage rises from 32.4% to 61.9% — it nearly doubles — and Washington DC’s from 52.0% to 94.7%. Melbourne barely moves, from 9.9% to 11.8%, because only 19% of its grid is buffer. The size of the shift is almost exactly the inverse of the “% inside” column, as expected for a fixed-width buffer applied to boundaries of very different sizes.

The *ordering*, by contrast, is largely preserved: the same three cities lead and the same three trail, with Washington DC rising from fourth to second the only substantial move. So the buffer distorts every absolute figure severely while leaving the comparative picture broadly intact. This is a strong argument for treating the coverage percentages in Table 5.2 as internal to this study’s grid definition rather than as statements about the cities themselves.

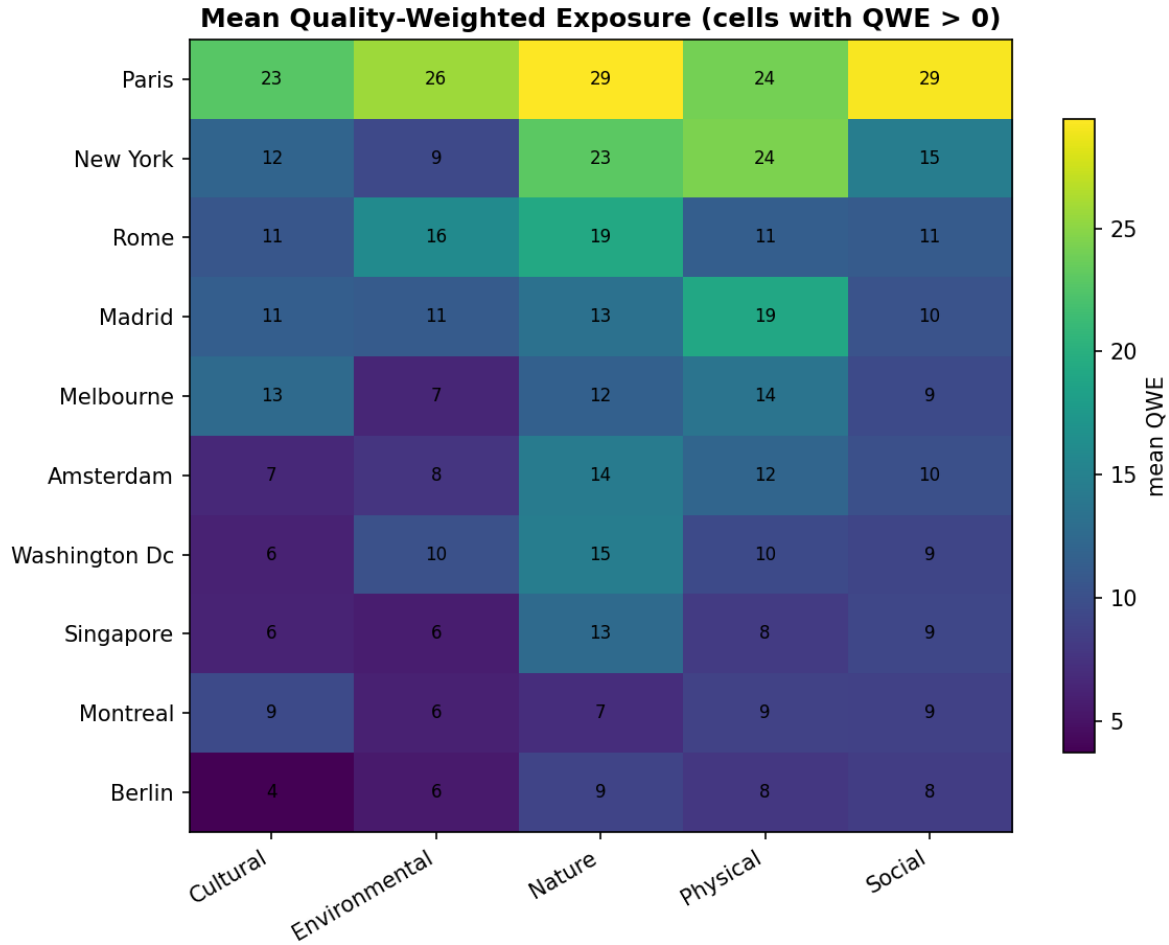


Figure 5.5: Mean Quality-Weighted Exposure across cells with QWE > 0, by city and dimension.

5.9.3 Gravity Decay Constant τ

Varying τ over $\{5, 10, 15, 20\}$ minutes leaves GMA *coverage* completely unchanged — 50.58% for Amsterdam, 73.76% for Washington DC and 37.77% for Madrid at every value tested.

This is not a null result to be passed over: it exposes that GMA coverage, as defined here, does not measure what its name suggests. Because the exponential decay $e^{-d/\tau}$ is strictly positive, every origin-destination pair inside the 30-minute search radius contributes a non-zero amount, so the count of cells with GMA > 0 is determined entirely by the 30-minute cutoff and not by τ at all. τ changes the magnitude of GMA and hence the relative ranking of cells within a city, but it cannot change how many cells are counted as covered. The GMA coverage column is therefore best read as “share of cells with any scored park within 30 minutes”.

5.9.4 QAMD Quality Threshold q

q is the most consequential of the four index parameters. Table 5.8 varies it from -0.5 to 1.0 standard deviations.

Coverage falls by a factor of three to four across the range: Amsterdam drops from 29.5% to 7.8%. More importantly, q can change the dimensional conclusion. Amsterdam and Madrid keep environmental as their worst-served dimension at every q , but Washington DC moves from nature ($q = -0.5$) to social ($q = 0$) to environmental ($q \geq 0.5$). Statements about which dimension a city

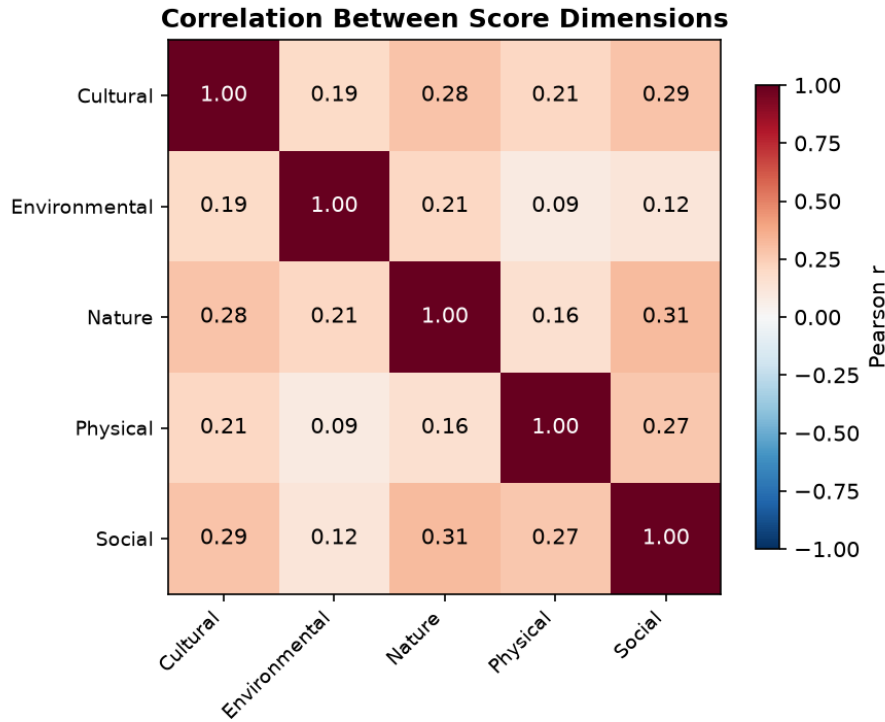


Figure 5.6: Pearson correlations between the five dimension scores across all parks in the 35-city Dietz sample. All off-diagonal values fall between 0.09 and 0.31.

under-provides are therefore conditional on where the quality bar is set, and should be reported with q stated explicitly.

5.9.5 Why Sweeps C and D Cover Three Cities

Sweeps A and B run over all ten cities; sweeps C and D over three (Amsterdam, Madrid and Washington DC). The reason differs for each, and only one of the two is a genuine limitation.

Sweeps A and B are cheap because they re-cut results the pipeline has already produced. QAMD stores the walking time to the nearest qualifying park, so varying the threshold T is a different comparison against an existing column, and the buffer sweep is a point-in-polygon test against cached cell centroids. Neither recomputes any distance.

Sweeps C and D cannot reuse anything. Both τ and q are applied *during* index computation rather than after it, so each value requires the origin-destination matrix to be rebuilt and the indices recomputed — for the four τ values and four q values together, roughly ten times the arithmetic of a single production run of the pipeline. The full OD matrix is not retained on disk for the larger cities (Melbourne’s runs to approximately 62 million rows), so it must be regenerated in memory chunk by chunk.

For τ this restriction costs nothing, because the invariance reported in Section 5.9 is algebraic rather than empirical. The gravity kernel $e^{-d_{ij}/\tau}$ is strictly positive for every finite τ , so every origin-destination pair inside the 30-minute search radius contributes a non-zero amount to GMA_i regardless of τ . The set of cells with $\text{GMA}_i > 0$ is therefore fixed by the search radius alone. The three cities confirm a result that holds by construction for all ten; additional cities would restate it, not test it.

For q the restriction is a real one. The finding that Washington DC’s worst-served dimension

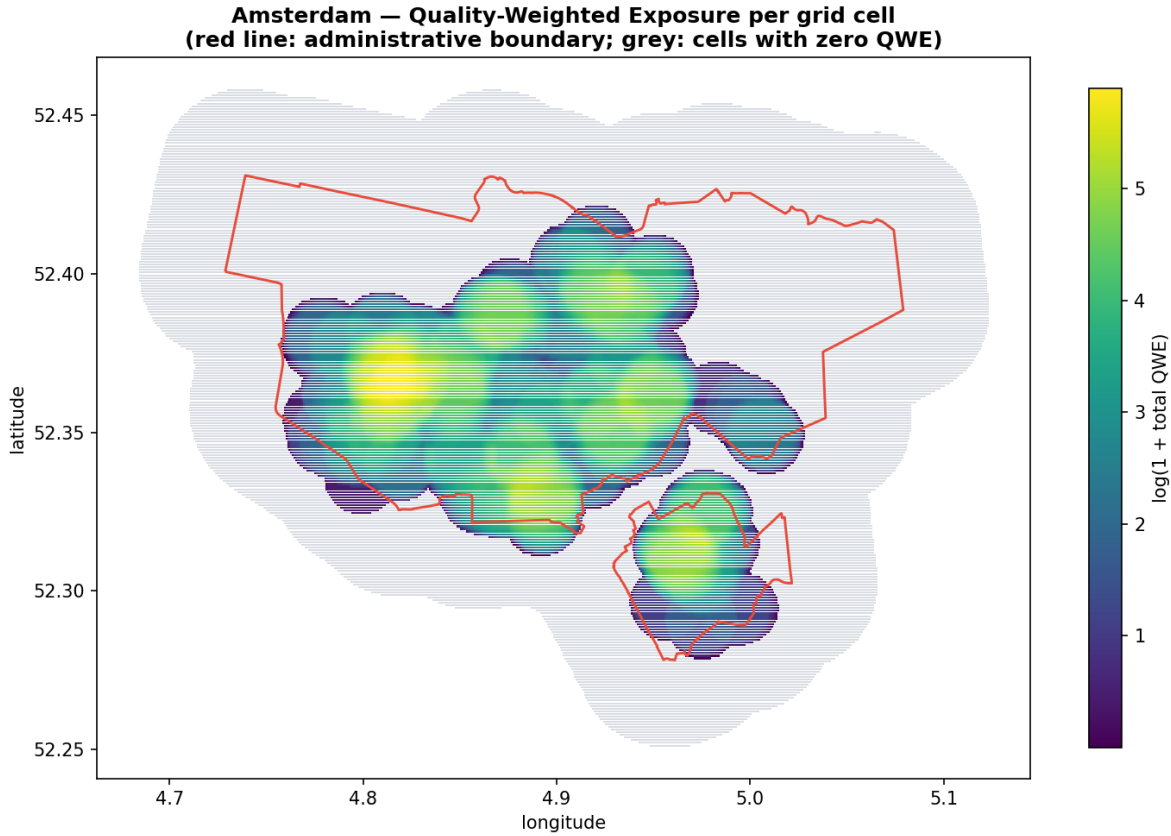


Figure 5.7: Total Quality-Weighted Exposure per grid cell, Amsterdam. Red line: administrative boundary; grey: cells with zero QWE. The unserved grey collar outside the boundary is the 3 km buffer that enters every coverage denominator (Section 4.9).

migrates from nature to social to environmental as q rises, while Amsterdam’s and Madrid’s remain environmental throughout, rests on a single city out of three. Whether q -sensitivity of this kind is common or exceptional across the full sample is not established here, and the q results should be read as indicative rather than as a characterisation of the sample. Extending the sweep is a matter of execution rather than method — the computation is unchanged, and because QAMD is a per-origin minimum, partitioning origins into chunks and concatenating the results is exact — so this is a tractable extension rather than an open problem.

The wider point is that the two parameters differ in kind. A conclusion about τ can be established analytically. A conclusion about q is empirical, and the confidence attached to it should scale with how much of the sample it was tested on.

5.9.6 Summary

Table 5.9 collects the verdict of each sweep against the claims made in Section 5.8, distinguishing those that survive the full parameter range from those that do not.

Table 5.7: Coverage with the published 3 km-buffered grid against the administrative area alone.

City	% inside	QWE %		GMA %		QAMD 15 %	
		buf.	admin	buf.	admin	buf.	admin
Paris	37.6	99.6	100.0	100.0	100.0	91.7	98.8
Montreal	48.3	61.8	89.6	80.3	97.7	49.3	75.2
New York	68.7	57.0	78.5	72.4	89.3	37.9	53.2
Washington DC	44.8	52.0	94.7	73.8	99.6	34.5	68.9
Berlin	61.2	47.8	73.3	66.1	88.8	33.1	51.9
Amsterdam	45.0	32.4	61.9	50.6	77.3	23.8	46.9
Madrid	55.5	27.5	47.8	37.8	60.5	20.6	36.2
Rome	63.1	25.5	39.8	38.8	58.2	16.1	25.0
Singapore	67.1	20.6	30.6	28.0	40.6	15.5	23.0
Melbourne	80.6	9.9	11.8	12.1	13.8	6.7	8.0

Table 5.8: Mean QAMD coverage (% of all cells, 15-minute threshold) as the quality cut q varies. $q = 0$ is the published setting.

City	$q = -0.5$	$q = 0$	$q = 0.5$	$q = 1.0$
Washington DC	47.8	34.5	23.9	14.7
Amsterdam	29.5	23.8	17.9	7.8
Madrid	24.4	20.6	13.7	10.0

Table 5.9: Robustness of the report’s main claims.

Claim	Verdict
City ordering by coverage	Robust. Rank correlation ≥ 0.976 across all thresholds; largely preserved when the buffer is removed.
Environmental is the weakest dimension	Robust to threshold (5/10 cities at every $T \geq 10$, all ten cities tested); holds under q variation in two of the three cities tested (Section 5.9.5).
Physical is the strongest dimension	Threshold-dependent. Seven cities at $T = 5$, five at $T \geq 15$.
Absolute coverage percentages	Not robust. Nearly double for compact cities once the 3 km buffer is removed.
GMA coverage	Insensitive to τ by construction , established analytically rather than empirically; determined by the 30-minute cutoff.
QAMD coverage	Highly sensitive to q ; varies three- to four-fold over a plausible range. Tested on three cities only (Section 5.9.5).

Chapter 6

Discussion

6.1 The Boundary Problem is the Main Result

The single most important outcome of extending this work from one city to ten is a negative one: the headline metric does not mean what it appears to mean. Reported in isolation, “32.4% of Amsterdam cells have quality-weighted green access” invites comparison with other cities. Table 5.2 shows that such comparisons are dominated by an artefact.

The mechanism is straightforward. The denominator of every coverage figure is the number of grid cells inside the city’s OSM administrative relation, and those relations follow no common convention. Paris’s relation stops at the Périphérique, excluding the dense inner suburbs; Melbourne’s extends across the entire metropolitan region including farmland. Cells in a rural fringe are counted in the denominator but can never be served, so a city is penalised for the administrative accident of having a generously drawn boundary.

This is a variant of the modifiable areal unit problem, well known in spatial analysis but easy to overlook when a pipeline is validated on a single city. It was not visible in the Amsterdam-only study, and would not have been visible in a two- or three-city one either — it took a sample with a 33-fold spread in grid size to make it unmistakable.

6.2 The Source Framework Already Guards Against This

Honesty requires a further point, which also identifies the cheapest fix. ATGreen does not report its indices over the whole buffered grid. Its index routine explicitly invalidates two classes of cell before anything is reported — those outside the administrative boundary (`inbound == 0`) and those with no population — and every downstream target-satisfaction calculation filters on `(inbound == 1) & (population > 0)`. ATGreen therefore uses the 3km buffer for the one thing it is good for, ensuring a park just outside the boundary still contributes to cells inside it, and then excludes the buffer ring and every unpopulated cell from the reported denominator.

The pipeline in this project inherited the buffer but neither filter. The artefact quantified above is consequently not a previously unrecognised problem in the field: it is the predictable result of dropping a guard the source framework already implements, and rediscovering it from the data is better read as a validation of ATGreen’s design than a critique of it.

Two things nevertheless remain contributions here. First, the magnitude is measured rather than assumed — $r = 0.93$ between coverage and park share, and the paired comparison in Table 5.7 showing Amsterdam’s coverage nearly doubling once the buffer ring is excluded. Neither figure is

published by ATGreen. Second, the diagnosis points to a cheaper remedy than the one usually reached for. Population weighting requires the GHS-POP raster, a separate multi-hundred-megabyte download that ATGreen's notebooks expect the user to supply and that this project never obtained; an `inbound` flag requires no additional data at all, only a point-in-polygon test against a boundary already computed in stage 1, and Table 5.7 shows it recovers most of the distortion by itself.

A second, compounding factor is the 3 km grid buffer documented in Section 4.9. The denominator of every coverage figure is the administrative area *plus* that buffer, and because the buffer is a fixed width rather than a proportion, it inflates small boundaries far more than large ones: 62% of the Paris grid lies outside the City of Paris, against 19% for Melbourne (Table 4.2). Its direction is opposite to the boundary-convention effect — it penalises compact cities — which means the two distortions do not simply add, and the net bias on any individual city is not straightforward to sign. Paris topping the table despite carrying the largest buffer penalty is, if anything, evidence that its lead is real.

Two mitigations are available. The first, adopted here, is to report a conditional measure (Table 5.4) alongside the raw one. The second, which would be preferable and is left to future work, is to weight each cell by its GHS-POP population count. The grid used in this project is already GHS-aligned, so the population raster can be joined directly; population weighting would drive the empty rural cells to near-zero weight automatically and yield a genuinely population-based accessibility measure rather than an area-based one. It is not, however, the cheapest: an `inbound` flag (Section 6.2) removes most of the same distortion using only the boundary polygon the pipeline already computes, and should be implemented first.

6.3 Quality Weighting vs. Unweighted Baselines

The gap between GMA and QWE coverage quantifies how many cells are nominally served by green space but only by low-quality parks. That gap is remarkably consistent across the sample: GMA exceeds QWE by 13–22 percentage points in nine of the ten cities. Paris is the exception, at 0.4 points, simply because both indices are saturated. The consistency suggests the quality filter is doing similar work everywhere rather than reflecting city-specific data quirks. The implication for practice is direct: unweighted coverage overstates useful access by a roughly constant 13–22 points, so a city meeting a green-access target on an unweighted measure may still leave a fifth of its nominally served cells with only low-quality provision.

6.4 Dimensional Findings Generalise

The Amsterdam study concluded that environmental accessibility was its weakest dimension. That conclusion survives the extension: environmental is worst-served in five of ten cities and best-served in none. Because the Dietz scores are z-standardised *within* each city, this is not an artefact of scoring — a city where environmental parks were merely as common as other types would show no systematic deficit. The pattern instead reflects that parks supporting gardening and active conservation of nature are genuinely rarer than parks with sports pitches, across very different planning traditions. The implication is a concrete planning lever: adding conservation and gardening features to existing well-located parks would raise environmental accessibility without acquiring any land, since the deficit is in park content, not park placement.

6.5 What “Above Average” Means Across Cities

The QAMD index counts a park as qualifying when its quality z-score is at or above zero. Because the Dietz scores are standardised within each city (Section 4.9), zero is that city’s own mean, not an international one. Roughly half of every city’s parks qualify by construction, whatever their absolute quality.

This has a specific consequence for the comparisons in this chapter: Table 5.4 does not show which city has the best parks. It shows how evenly each city distributes its own better-than-median parks relative to where its grid cells are. Those are different questions, and the second is arguably the more useful one for planning — it identifies internal inequity rather than ranking cities against each other — but the distinction must be kept in view. An absolute cross-city quality comparison would require re-standardising the Dietz fingerprints against a pooled global distribution, which the published score files do not support.

6.6 Geodesic vs. Street-Network Distances

As in the single-city study, all distances are straight-line haversine converted at 5 km/h. Real walking routes are 20–30% longer, so every coverage figure reported here is an optimistic bound. This bias is not uniform across cities: it is worst where street networks are least permeable to pedestrians, which in this sample means the North American cities with large blocks and limited crossings, and mildest in the dense European cores. Comparisons between Paris and New York in particular should be treated as indicative rather than precise until OSRM routing is integrated.

6.7 Data Quality

Two data issues emerged that were invisible at single-city scale. First, park recovery from the OSM extracts averaged 94.3% but fell to 71.7% for Washington DC, where a tightly clipped extract excludes parks lying just outside the federal district. Coverage figures for DC are therefore understated relative to the others. Second, four parks across Singapore, Paris and New York had invalid geometries — self-touching rings produced when multipolygon relations are assembled — which caused the geometric intersection step to fail outright. These were repaired with a zero-width buffer, the same operation the pipeline already applies to simple park ways. Neither issue arises in Amsterdam, which is precisely why extending to a larger sample was necessary to find them.

6.8 Limitations

- **No population weighting.** Coverage is area-based, not population-based. This is the principal limitation and the direct cause of the boundary problem discussed above.
- **The grid includes a 3 km buffer** beyond each administrative boundary, accounting for 19–62% of the denominator depending on city size (Table 4.2). ATGreen applies the same buffer but excludes the ring from reporting via an `inbound` flag (Section 6.2); this pipeline did not, which is the direct cause of the artefact in Section 5.3.
- **Quality is relative to each city.** Within-city z-standardisation means “above average” is city-specific; no absolute cross-city quality claim is supported.
- **Index parameters are uncalibrated.** $\tau = 10$ min, $q = 0$ and the 15-minute threshold are convention rather than derived from these data. Section 5.9 quantifies the consequences: the

threshold barely affects the city ranking, but q moves QAMD coverage three- to four-fold.

- **Uniform 5 km/h walking speed** for all people, ignoring gradient, barriers, age and mobility.
- **Straight-line distances.** Coverage is systematically overstated, unevenly across cities.
- **Boundary non-equivalence.** The ten relations are not comparable spatial units.
- **Park recovery is incomplete**, and unevenly so.
- **Ten cities, not thirty-five.** The sample is deliberately diverse but small, and the correlations reported here rest on ten observations.
- **The q sensitivity sweep covers three cities.** Unlike the threshold and buffer sweeps, it requires rebuilding the origin-destination matrix per parameter value; see Section 5.9.5. Its findings are indicative rather than sample-wide.

Chapter 7

Legal, Social, Ethical and Professional Issues

This project processes only public, aggregated geospatial data and involves no human participants, so it raises no ethical-approval requirement. It is not, however, free of legal and social consequence: it redistributes licensed third-party data, and it produces numbers that could be read as a ranking of cities’ green provision. This chapter sets out the obligations that follow and how they were discharged.

7.1 Legal Issues

7.1.1 Data Licensing

Three external datasets are used, each under a distinct licence with distinct obligations (Table 7.1).

Table 7.1: Licensing of the external datasets and the obligations each imposes.

Dataset	Licence	Obligation
OpenStreetMap (via Geofabrik) [19]	ODbL 1.0 [23]	Attribute “© OpenStreetMap contributors”; share-alike applies to any redistributed <i>derived database</i>
GHS-POP grid definition [3]	CC BY 4.0 [25]	Credit the European Commission JRC; indicate if changes were made
Dietz et al. park scores [2]	Academic replication package	Cite the source publication; use for research

The ODbL [23] is the obligation requiring most care, because it distinguishes a *Produced Work* from a *Derivative Database*. Statistical outputs such as the coverage percentages, figures and tables in this report are Produced Works: they require attribution but do not trigger share-alike. The per-cell CSV files and the extracted `parks_<city>.geojson` files are different in kind — they contain OSM-derived geometry and would constitute a Derivative Database if published, and would then have to be released under the ODbL itself. This distinction is recorded here so that any future release of the intermediate data is made on the correct terms rather than by oversight.

7.1.2 Data Protection

No personal data is processed at any stage. The finest spatial unit is a 3 arc-second grid cell of roughly 90 m, which is an aggregate containing no individual-level attribute, and no data subject is identified or identifiable at any point in the pipeline. The work is therefore outside the material scope of the UK GDPR and the Data Protection Act 2018 [27], and no data protection impact assessment is required.

It is worth noting that the upstream provider has already applied a data-minimisation step. Geofabrik's published extracts deliberately omit the user names, user IDs and changeset IDs of OSM objects, on the stated basis that these fields are assumed to contain personal information about OpenStreetMap contributors and are therefore subject to EU data protection regulation. The pipeline consumes these reduced extracts and never requests the full-metadata versions, so contributor identities are never handled.

7.1.3 Software Licensing and Attribution

The implementation depends on `numpy`, `pandas`, `geopandas`, `shapely`, `pyproj`, `pyosmium`, `scipy`, `matplotlib`, `rtree` and `pyarrow`. All are distributed under permissive licences (BSD, MIT, Apache 2.0 or equivalent) that allow use, modification and redistribution in academic work provided copyright notices are preserved. None imposes a copyleft obligation on the code written for this project.

The delineation between original and third-party work is stated explicitly. The accessibility indices, the pipeline and the analysis code are original. The park quality scoring is the work of Dietz et al.; the grid-based accessibility framework is the work of Battiston and Schifanella. Both are cited wherever their contribution is used, and no third-party code has been incorporated without attribution.

7.2 Social Issues

Access to green space is a determinant of public health, and it is unevenly distributed. Work of this kind is therefore not neutral: it produces figures that could inform where investment goes. Three risks follow directly, and each is a reason for a specific choice made in this report.

7.2.1 The Risk of a Misleading League Table

The clearest social risk is created by this project's own headline metric. Section 5.3 shows that raw coverage percentages correlate at $r = 0.93$ with the share of a city's grid occupied by parks — a quantity determined largely by where an administrative boundary was drawn. Published without that caveat, Table 5.2 would read as a ranking in which Melbourne provides a tenth of the green access of Paris. A planning authority acting on that reading could misdirect investment on the strength of a cartographic artefact.

This is why the conditional measure in Table 5.4 is reported alongside the raw figures rather than instead of them, why the sensitivity analysis in Section 5.9 is included, and why Section 4.9 states the boundary and buffer assumptions explicitly. Presenting a metric together with the evidence of where it fails is a social obligation as much as a methodological one.

7.2.2 Mapping Bias and the Risk of Entrenching Inequity

Park quality here is inferred entirely from OpenStreetMap tags, and OSM is produced by volunteers whose density and thoroughness are not uniform. Haklay's [29] comparison of OpenStreetMap

against Ordnance Survey found completeness varying systematically with area, and specifically that coverage was poorer in more deprived districts. Amenity mapping tends to be richer in affluent and well-populated areas than in deprived ones. A park in an under-mapped neighbourhood can therefore score poorly not because it lacks benches, play equipment or trees, but because nobody has recorded them.

The consequence is a fairness concern with a specific direction: if these indices were used to allocate resources, under-mapped areas would appear less well served in a way that reflects mapping effort rather than provision, and the correction would flow to places that are already better documented. This is stated as a limitation rather than solved, and any operational use of these outputs should be preceded by ground-truthing in the areas identified as poorly served.

7.2.3 Green Gentrification

Improving park provision is not costless to existing residents. Wolch et al. [28] document a pattern in which green-space investment raises surrounding property values and rents, displacing the lower-income residents the investment was intended to benefit — the paradox they summarise as the need to make cities only ‘just green enough’. A tool that identifies underserved neighbourhoods identifies, by the same token, targets for property development. This project cannot resolve that tension, but it should not be presented as if the tension does not exist: the appropriate use of these indices is alongside housing policy, not as a standalone investment prospectus.

7.3 Ethical Issues

Human participants. None. No survey, interview, observation or personal data collection was undertaken, so no ethics review was required under the College’s research ethics framework.

Honest reporting of negative results. The most substantial ethical demand in a project of this type is the willingness to publish findings that undermine one’s own headline result. The boundary-artefact finding in Section 5.3, the demonstration that GMA coverage is mathematically independent of its own decay parameter, and the sensitivity analysis showing that the “physical is strongest” claim is threshold-dependent all reduce the apparent strength of the contribution. Each is reported in full, because a reader cannot calibrate confidence in results whose fragilities are concealed.

Avoiding overclaim. Because the Dietz scores are standardised within each city, the indices cannot support absolute cross-city quality comparison, and the report states this rather than allowing the tables to imply otherwise (Section 4.9). Similarly, coverage figures are described as optimistic bounds, since straight-line distances understate real walking routes.

Scope of inference. The results describe modelled geometric accessibility. They do not measure park use, perceived safety, or health outcomes, and no causal claim about well-being is made from them.

7.4 Professional Issues

The project was conducted with reference to the BCS Code of Conduct [26], whose four principles map onto the work as follows.

Public interest

Due regard was given to public health as the motivating concern, and to the legitimate rights of third parties — the OpenStreetMap contributors whose volunteered work underpins the

entire analysis, and whose licence terms are respected. The social risks in §7 are documented rather than left implicit.

Professional competence and integrity

Work was kept within the limits of demonstrated competence, and the boundaries of that competence are stated. Where the implementation falls short of the intended design — geodesic rather than street-network distances, unweighted rather than population-weighted cells — this is reported as a limitation rather than obscured. No claim is made beyond what the evidence supports.

Duty to relevant authority

Supervisory guidance was followed, the work is the author's own except where explicitly attributed, and academic integrity requirements on plagiarism and referencing have been observed throughout.

Duty to the profession

Reproducibility was treated as a professional obligation, not an optional extra. Every parameter is documented (Section 4.9), the pipeline runs offline from archived inputs so results are reproducible without external services, and outputs were verified against the previously published single-city figures to confirm the extension did not silently alter the baseline.

Two further practices support this. Intermediate results are cached and verified rather than trusted: the chunked computation was checked to produce byte-identical output at different partition sizes, and the accelerated extraction routines were validated against the original implementations before being relied upon. And the limitations chapter is written to be actionable — each limitation names what would be required to remove it, so that a subsequent worker inherits an honest account of what remains to be done.

Chapter 8

Conclusion

This project extended a quality-weighted green accessibility pipeline from a single city to ten, spanning five continents and 3.3 million grid cells. The four indices — QWE, QAMD, GMA and BEST — behave consistently everywhere, preserving the expected ordering $GMA > QWE > QAMD_{15}$ in all ten cities, and Amsterdam’s original figures are reproduced exactly.

The substantive findings are twofold. Dimensionally, the environmental dimension is the most underprovided across the sample, worst-served in five of ten cities and best in none, while the physical dimension is the best-served; the conclusion drawn from Amsterdam alone therefore generalises. Methodologically — and more importantly — raw coverage percentages are not comparable across cities. They correlate at $r = 0.93$ with the share of the delimited area occupied by parks, which is a consequence of administrative boundary conventions rather than of urban green provision. Conditioning on cells that can reach a park at all compresses the apparent ten-fold spread to less than two-fold and materially reorders the cities.

The practical recommendation that follows is that multi-city green accessibility work should either weight by population or report conditional measures, and should state explicitly which administrative unit defines each city. A ranking built on unconditioned coverage percentages measures cartography at least as much as it measures parks.

It should be said plainly that ATGreen already applies this discipline, reporting its indices only for cells that are inside the boundary and populated (Section 6.2). The contribution here is not the recognition that such a guard is needed, but the measurement of what its absence costs — and the observation that the cheaper half of that guard requires no data beyond what this pipeline already computes.

8.1 Future Work

Seven extensions follow directly from the limitations set out in Section 6.8, ordered here by the ratio of expected benefit to implementation cost.

1. **Apply the inbound flag.** The single cheapest correction. The 3 km buffer ring contributes between 19% and 62% of the denominator depending on city (Table 4.2), and every cell in it is already tagged with the boundary polygon the pipeline computes anyway. Excluding the ring from reporting — while retaining it for distance computation, so that edge cells still see parks just outside the boundary — would remove the larger part of the artefact documented in Section 5.3 for perhaps a day of work.
2. **Weight cells by population.** The structurally correct fix, and the one that would make

coverage figures mean what readers assume they mean. Because the grid is generated to the GHS-POP convention rather than approximated (Section 4.3), the population raster joins cell-for-cell with no resampling. Population weighting drives empty rural cells to near-zero weight automatically, converts every index from an area-based to a person-based measure, and would make the conditional measure proposed in Section 5.4 unnecessary rather than merely useful.

3. **Replace geodesic distances with street-network routing.** Haversine distances ignore rivers, rail corridors, motorways and gated blocks, and therefore overstate accessibility — unevenly, since the size of the error depends on how severed a city’s street network is. Routing through OSRM, as ATGreen does, would both correct the bias and quantify it per city; the ratio of network to straight-line distance is itself an interesting cross-city statistic and is not reported anywhere in this literature.
4. **Extend to the full 35-city sample.** The correlation of $r = 0.93$ between coverage and park-area share rests on ten observations. Thirty-five would allow the boundary effect to be estimated rather than merely demonstrated, and would permit the cities to be grouped by boundary convention — municipality, city-state, metropolitan region — and compared within group.
5. **Complete the q sensitivity sweep.** The quality-threshold sweep covers three cities rather than ten because each value of q requires the origin–destination matrix to be rebuilt (Section 5.9.5), and q is the parameter to which the results are most sensitive: it moves QAMD coverage three- to four-fold. Caching the matrix once and re-filtering it per threshold, rather than recomputing, would make the full sweep affordable.
6. **Standardise quality scores globally rather than within city.** Because the Dietz scores are z-standardised inside each city, “above average” means something different in Paris than in Melbourne, and no absolute cross-city quality claim is licensed by the present results. Refitting the log–log residual model on the pooled sample would produce comparable scores, at the cost of making each city’s scores dependent on the composition of the sample.
7. **Repair park recovery for clipped extracts.** Recovery averaged 94.3% but fell to 71.7% for Washington DC, where the Geofabrik extract is clipped tightly to the federal district and excludes parks a few hundred metres outside it. Reading DC from a wider mid-Atlantic extract would close the gap; the same check should be run for any city whose extract boundary sits close to its administrative boundary.

Two further items are worth recording as open questions rather than tasks: the index parameters τ , q and T are conventions rather than calibrated quantities, and calibrating them would require observed park-visit data that does not exist for this sample; and the five well-being dimensions are treated as equally important, which is a value judgement that a participatory study, not a computational one, would have to settle.

8.2 Lessons Learned

Check the denominator before refining the numerator. The greatest part of the effort in this project went into the four indices — their formulation, their normalisation, the handling of ties and missing values. The finding that eventually mattered was that the denominator, the set of cells over which coverage is averaged, was not comparable across cities. A single scatter plot of coverage against park-area share, which took minutes to produce, explained more of the cross-city variation ($r = 0.93$) than any refinement of the indices did. Were the project to be started again, that diagnostic would

come first: before comparing any two cities on any measure, establish that the units being compared are the same kind of thing.

Read the reference implementation, not only the paper. ATGreen's published description does not dwell on its inbound flag; the code does. Several weeks of interpreting the Melbourne result as a substantive finding about Australian urban form would have been avoided by reading the source framework's filtering logic at the start rather than at the point of writing up. Where a project extends an existing implementation, the implementation is part of the literature.

A larger sample is a debugging tool, not only a generalisation claim. Two defects were invisible at single-city scale and immediate at ten: four parks with self-touching multipolygon rings, which caused the geometric intersection step to fail outright for Singapore, Paris and New York, and the clipped Washington DC extract. Neither occurs in Amsterdam. Extending the sample was undertaken to test whether the findings generalised, but its more immediate value was exposing failure modes that a single well-behaved city had concealed.

Design for the memory ceiling early. The decision to stream the PBF files in two targeted passes, resolving coordinates only for the park identifiers actually needed, was made after an initial approach exhausted memory on the Spanish extract. Retrofitting it required reworking the extraction stage rather than adjusting it. Country-sized inputs were foreseeable from the start, and a design that assumed them would have cost less than the one that had to be rescued.

Build sensitivity analysis in from the beginning. The sweeps over T , the buffer, τ and q were added late, and the cost of that timing is visible in the results: the q sweep covers three cities rather than ten because, by the time it was written, the pipeline had not been structured to reuse the origin–destination matrix across parameter values. Had parameters been treated as sweepable from the first version, the full sweep would have been nearly free.

Reproducibility is cheaper to maintain than to reconstruct. Committing to local PBF extracts with recorded snapshot dates, rather than live Overpass queries, made every result in this report re-derivable months later and made the temporal-consistency discussion in Section 3.3.3 possible at all. The discipline felt like an overhead at the time and was repaid the first time a number had to be explained.

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Appendix A

Reproducibility

A.1 Environment

The pipeline runs on Python 3.11 or later. Table A.1 lists every third-party package required, with the version used to produce the results in this report.

Table A.1: Python dependencies and versions.

Package	Version	Purpose
numpy	2.4.6	Vectorised distance computation
pandas	3.0.3	DataFrames and CSV I/O
geopandas	1.1.3	Spatial DataFrames and spatial joins
shapely	2.1.2	Geometry operations (Polygon, intersection)
osmium	4.3.1	OSM PBF parsing
pyproj	3.7.2	CRS transformations (UTM projection)
matplotlib	3.11.0	Visualisation
scipy	1.18.0	Scientific computing
scikit-learn	1.9.0	Classification utilities
tqdm	4.68.3	Progress bars
requests	2.34.2	Overpass API (alternative pipeline)
rtree	1.4.1	Spatial indexing

Python version: 3.14.0 Virtual environment: .venv/

A.2 Running the Pipeline

```
1 cd "Final project"
2 .venv\Scripts\activate           # Windows
3 # source .venv-linux/bin/activate # Linux
4
5 # One city at a time, using the staged runner
6 python "KCL Main Project/Execution/run_city_staged.py" ^
7     amsterdam pbf_data/noord-holland-260620.osm.pbf ^
8     --out-dir output_figures --chunk-origins 12000
9
```

```

10 # ... repeated for berlin, madrid, melbourne, montreal, new_york,
11 #    paris, rome, singapore, washington_dc
12
13 # Per-city figures (one generated copy per city)
14 python "KCL Main Project/Execution/visualize_scores_paris.py"
15
16 # Cross-city comparison
17 python "KCL Main Project/Execution/crosscity_compare.py" --out-dir
    output_figures

```

Listing A.1: Commands to reproduce the ten-city results.

Every stage is cached, so an interrupted run resumes rather than restarting. The `-chunk-origins` flag bounds peak memory by processing origin cells in blocks; results are invariant to its value, which was verified by running Amsterdam at two different chunk sizes and comparing the output byte-for-byte.

A.3 Supporting Modules

The original `quality_accessibility_osmium.py` is unmodified. Four additive modules rebind functions at runtime to make the ten-city run tractable; each was verified against the original before use, as recorded in Table A.2.

Table A.2: Additive modules and their verification.

Module	Purpose and verification
<code>run_city_staged.py</code>	Resumable seven-stage wrapper with origin chunking. Chunk-invariance verified byte-for-byte on Amsterdam.
<code>fast_grid.py</code>	Vectorised grid construction. Reproduces Amsterdam's 92,655 cells with identical geometries, $\approx 10\times$ faster.
<code>fast_distance.py</code>	KD-tree distance matrix. Reproduces Amsterdam's 9,072,560 OD pairs exactly, same pair set and distances.
<code>targeted_parks.py</code>	Resolves node coordinates only for known target park IDs rather than indexing every node. Required for the province- and country-sized extracts (Quebec 1.1 GB, Spain 1.4 GB), which exhaust memory otherwise. Verified identical on Amsterdam (106 parks) and Melbourne (1,787 parks).

Appendix B

Generated Output Files

For each of the ten cities <city>, the pipeline writes the files listed in Table B.1 to `output_figures/`:

Table B.1: Per-city and cross-city outputs.

File	Description
<code>qwe_<city>.csv</code>	Quality-Weighted Exposure per cell
<code>qamd_<city>.csv</code>	Quality-Adjusted Minimum Distance per cell
<code>gma_<city>.csv</code>	Gravity-Model Accessibility per cell
<code>best_<city>.csv</code>	Best Reachable Score per cell (5/15/30 min)
<code>cell_scores_<city>.csv</code>	Area-weighted quality scores per green cell
<code>overlap_<city>.csv</code>	Park-cell intersection areas
<code>cells_<city>.csv</code>	Cell centroids
<code>meta_<city>.csv</code>	Grid size and green-cell count
<code>parks_<city>.geojson</code>	Extracted park geometries
<code><city>_qwe_histogram.png</code>	QWE distribution per dimension
<code><city>_qamd_histogram.png</code>	QAMD distribution with 15-min reference line
<code><city>_gma_histogram.png</code>	GMA distribution per dimension
<code><city>_coverage.png</code>	Coverage bar chart across indices
<code>radar_<city>.png</code>	City profile vs the 35-city average
<code><city>_index_summary.csv</code>	Summary statistics for all indices
<code>crosscity_summary.csv</code>	One row per city $\times$ dimension
<code>crosscity_coverage.png</code>	Coverage by city and index
<code>crosscity_qamd_15min.png</code>	City $\times$ dimension heatmap
<code>crosscity_qwe_mean.png</code>	Mean QWE by city and dimension
<code>crosscity_dimension_gap.png</code>	Within-city dimensional inequality